Interrupting the Loop: Periodic Subject Changes Raise Judged Surprise and Connection in Base Language Models

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Abstract

Where does the novelty a language model can produce with no task come from? We test the three places usually credited, the sampler, the prompt and the loop, on base models, with an LLM judge that is itself measured. An entropy-banded anti-probable decoder roughly doubles n-gram novelty against the training corpus but shows no detectable effect on judged surprise or connection; input improbability shows no benefit under the operationalizations we tested. The loop is different. Forced to continue a premise with the end-of-text token masked, a base model degenerates into repetitive modes; a closed loop built on the architecture of spontaneous cognition revives it, and taking that loop apart over 23 conditions locates most of the effect in two simple operations: a windowed repetition penalty (habituation) and a periodic *interruption*, one neutral sentence injected every few hundred tokens. On windows of generated text only, judged by Claude Opus 5 ($k = 5$) with the premise as the unit ($n = 10$), the interruption raises surprise from 1.6 to 3.0 and connection from 1.3 to 3.7 over habituation alone (paired permutation $p = 0.002$) and matches the full scaffold. Part of that is self-copy: given the same four sentences in rotation, the model replays its own earlier segments, out of the judge's view; on fresh windows the interruption adds +1.2 to +1.4 surprise and +0.8 connection, and a reset context, which cannot replay, adds +1.9 and +2.0. Controls show what carries the effect: a bare paragraph break does nothing and a continuity connective hurts; the new subject carries most of it; injecting the premise or the stream's own past is as bad as not interrupting; salience timing is no better than a clock. The gains are local: judged as whole documents, no arm builds an integrated text. A pre-registered replication on ten new premises confirms the primary contrast ($p = 0.003$) and refutes the expectation that the kept context carries connection. The ordering replicates on three base models from two families, under a second judge family and in the rankings of independent human readers. These results characterize a simple, controllable intervention for forced open-ended generation; they do not establish a general mechanism of creativity.

1 Introduction

Ask a language model for something new and you will usually get something fluent, plausible and familiar. Three places are commonly blamed, and three places are commonly optimized: the *sampler* (the tail of the next-token distribution, tamed by nucleus, typical or min-p truncation, or courted by higher temperature), the *prompt* (the input, engineered to steer the model somewhere it would not go on its own), and, less often, the *loop*: what happens when a model's output becomes its own next input and nobody asks anything.

The loop is where human novelty is usually placed. Insight rarely arrives as the answer to a strange question; it arises inside a closed circuit of thought feeding on its own output, during incubation, mind-wandering and sleep, when a spontaneously generated deviation survives critical review and is linked back to what came before. The neuroscience of creative cognition describes three coupled systems: a default-mode network that generates, an executive network that evaluates, and a salience network that decides what deserves attention; more creative people show more coupling between the first two, with salience regions coupling first [2, 3]. Creativity itself is often characterized as connecting semantically distant concepts [14], and incubation, leaving a problem and coming back to it, has a measured effect on solving it [27].

This paper asks, with controls, where the novelty a base language model can produce with no task comes from, and which of the operations usually credited for it survive measurement. The outcome we measure is narrow and we name it as such: *judged narrative surprise, connection and coherence* on short windows of forced open-ended continuation, as read by an LLM judge that is itself measured (repeatability, a second judge family, human readers). It is one ingredient of creativity, the appropriately unexpected turn, and not creativity; the title names the question the program set out with, and the results answer this narrower one. We test the three places in turn. They are three related studies (different generators, tasks and sample sizes, joined by one question and one measurement philosophy), not a factorial decomposition of sampler, prompt and loop under common conditions; the loop study is the paper's contribution and the other two are its motivation. One measurement stack is shared by every experiment: base models run locally at 8 bits (a scope decision; Section 3); LLM judges from a different model family than the generator, sampled k times per window with the median as the score; the premise, not the window, as the unit of inference in the loop experiments; and, where a public training corpus exists, verbatim novelty computed against it.

The sampler. We build the strongest version of the "fertile error" idea we could: an entropy-banded anti-probable decoder that, at the steps where the model is genuinely undecided, rescores the plausible candidates by their semantic distance from the recent context, with a coherence floor that never admits an implausible token. It works exactly as far as the surface: 4-gram novelty against the OLMo-2 training corpus roughly doubles and verbatim 8+-word training blocks fall from 0.80 to 0.20 of generations, at a small perplexity cost. But inside a reverie loop the same push shows no detectable difference from plain sampling in judged surprise, connection or novel delta, and in verified search (bin packing, two models, 3,200 verified heuristics) it does not separate from plain sampling either. Surface novelty is real and, at the present resolution, does not climb: a mechanistic counterpart of the finding that most n-gram-novel expressions are not judged creative [26].

The prompt. We build inputs that sit far from any human prompt (distant concept pairs, composed improbable contexts; distance measured against 10,000 real prompts) and let a strong model develop them. A typical request matches or beats every improbable arm; input improbability does not correlate with judged novelty. We found no benefit from prompt improbability under the tested operationalizations.

The loop. Left to continue a premise with the end-of-text token masked, a base model degenerates within a few hundred tokens into repetitive or corpus-like modes (literal orbits, website footers, exam keys, translation tables) and stays there. A closed reverie loop built on the architecture of spontaneous cognition (a salience monitor over the stream's own telemetry gating review, selective forgetting, reseeding, re-encounter with the premise) revives it. Then we take the loop apart, in three batteries and 22 conditions, and find that most of its effect lives in two operations, neither of them the elaborate ones: *habituation*, a windowed repetition penalty that keeps the loop from eating its literal past, and *interruption*, a new starting sentence injected every few hundred tokens over the preserved context. It is the machine equivalent of getting up for a coffee: leaving a thread one has been developing and being made to start somewhere else.

We show what a good interruption is made of: it must lead away (injecting the premise or the stream's own past is as bad as not interrupting), a boundary without a new subject does nothing and a boundary that asks for continuity hurts, the new subject carries most of the effect and habituation adds to it, it must be new each time (a fixed rotation of four sentences makes the model replay its own earlier segments, which a window judge cannot see), and it works whether the earlier text is kept in the model's context or dropped (dropped scores higher, in part because it cannot replay), a clock is as good a metronome as the salience monitor at the same rate, and no period beats a break every 150–300 tokens on the generated text itself. We also show what the interruption does not do: read as whole documents, none of these streams builds an integrated or developing text, and the interrupted stream reads as a sequence of restarts; the operator acts on windows, not on wholes. The ladder replicates on three generator models from two families and under a second judge family, its ordering is reproduced by independent human readers, and a pre-registered replication on ten new premises confirms the primary contrast. Finally we look inside the network, descriptively: residual-stream geometry at 13 sampled layers shows that bare generation moves least at every sampled layer, that judged surprise co-varies with surface departure over deep continuity, and that the interruption that scores best barely moves the deep state, whereas the scaffold's forgetting reseed moves it toward the beginning.

Contributions.

1. A calibrated anti-probable decoder and the measurement that its novelty is real at the surface (objective, against the training corpus) and undetectable at the level of ideas (Section 4); a null on improbable inputs, with input improbability measured (Section 5).
2. A reverie loop with a salience monitor and selective forgetting, and its dismantling into two operations, habituation and interruption, with a factorial and control battery that shows which part of an interruption carries the effect and characterizes its content, timing and period; the ladder replicated on three generator models from two families (Section 6).
3. An instrument section that measures the LLM judge before using it and checks it against a second judge family and against independent human readers, with a seed-level analysis and generated-only windows adopted after an external review (Sections 3, 7).
4. A descriptive residual-stream analysis relating judged surprise to what the network does at each sampled depth, and a dissociation between deep representational change and judged quality (Section 8).

Code, per-run data, judgments, human-rating packs and the dated laboratory notebook are released with the paper.

2 Related work

Decoding and the tail. Nucleus sampling [13] truncates the unreliable tail of the next-token distribution; locally typical sampling [18] targets human-like surprisal; min-p [21] scales the truncation with the model's confidence and is exactly the coherence floor we use; contrastive decoding [16] and plug-and-play steering [7] shape the distribution toward or away from a reference. All of these regulate the tail. Our anti-probable sampler *courts* it, inside an entropy band and above a floor, and we find that this buys surface novelty only.

Repetition, self-reinforcement and entrainment. The collapse of a base model left to itself is a well-studied phenomenon. Zhu et al. [30] name the *self-reinforcement effect* (a repeated token becomes more probable each time it repeats) and suppress it with a repetition penalty restricted to a recent window; the graded, windowed penalty we call *habituation* is a decoding-time operator of the same family, and we claim no novelty for it. Guan and Huang [10] attribute the tendency to repeat to a learning bias of maximum-likelihood training and correct it with self-contrastive training; Xu et al. [29] track the distance between the current next-token distribution and past ones to detect repetition and topic drift and steer decoding accordingly. At the mechanistic level, Niu et al. [22] show that language models assign higher probability to tokens present in their context even when those tokens are irrelevant (*contextual entrainment*), and locate heads that carry it. Our bare arm is entrainment on the model's own output; what this paper adds is not a new remedy for repetition but a measurement of what the remedy leaves untouched (a habituated stream is fluent and still not creative) and of what a second, structural operator adds on top of it.

Evaluating creativity, and the judge. Nakajima et al. [20] show that the widely used Divergent Association Task ignores appropriateness and propose to score novelty conditional on it; our three separate dimensions (surprise, connection, coherence), reported without a composite, follow the same logic of not letting novelty be bought with incoherence. Saakyan et al. [26] show with 8,618 expert annotations that ≈91% of top-quartile n-gram-novel expressions are not judged creative; our factual-paraphrase escape mode and the sampler's idea-level null are independent confirmations of the same gap. Infini-gram [17] and Rusty-DAWG [19] make verbatim novelty against a training corpus computable at scale; we use infini-gram against the OLMo-2 corpus. On the instrument itself, Fein et al. [9] find that the strongest off-the-shelf judge they tested agrees with human preferences on creative-writing pairs only 73% of the time, and Haldar and Hockenmaier [11] document low intra-rater reliability of LLM judges across runs. Both cautions apply to this study. We answer the second by scoring every window with $k = 5$ independent calls and using the median, and by reporting the resolution of the instrument; we answer the first only partly, with a second judge family and with human raters on a subset, and we treat the judge as the study's main limitation (Section 10).

Loops and steering in story generation. Outer loops that steer a generator are common in story generation: SWAG [24] lets a second model choose the next action for the story model; Collective Critics [1] refine a plan and its expression with several critic models; STORYTELLER [15] adds a plot-planning structure to keep long stories coherent and cohesive. Against this literature our contribution is not the idea of an outer loop but its reduction: the operator we isolate is one injected sentence on a preserved context, without a planner, a critic or a reward; the factorial ablation that shows which part of the loop carries the effect; and the temporal characterization (period, decay, timing) that a planning framework does not provide.

Verified search. FunSearch [25] and AlphaEvolve [23] pair a language model with a hard evaluator and let selection do the work. Our bin-packing null (the anti-probable operator does not separate from plain sampling under verified evolution) and the loop results suggest that the structure of the generation loop, not the sampling operator, is where to look next.

Reading the residual stream. The logit lens is known to be fragile in intermediate layers, and the tuned lens [4] was proposed as a less biased alternative. We use the logit lens only for a coarse commitment layer and otherwise work with mean-centered residual geometry; the network section is descriptive, and its readings should be repeated with a tuned lens before they are taken as mechanism.

Creative cognition. Creative thought as coupling of default-mode and executive networks [2], with salience regions coupling first [3]; creativity as connecting semantically distant concepts [14];

incubation effects [27]; predictive processing [6]; dreams as anti-overfitting noise [12]; Boden's novelty/surprise/value triad [5]; conceptual blending [8]; novelty search [28]. The reverie loop of Section 6 was an explicit engineering of the first three into a text loop; its ablation is, in a sense, a test of which of these ingredients a language model needs, and the answer (an interruption on a clock) is humbler than the hypothesis.

3 Method

3.1 Measurement stack

Every experiment shares one stack. **Generators** are three base models from two families: Qwen3-8B-Base and Qwen3-30B-A3B-Base (a mixture-of-experts model with 3B active parameters, 53 tokens/s, the main generator), and OLMo-2-13B (whose training corpus is public). All are quantized to 8 bits (MLX affine quantization, group size 64) and run on Apple silicon through MLX with a numpy sampler that sees the full logits. Restricting to base models is a scope decision, not a finding: an instruction-tuned model brings a dialogue format and preference training that change what an open-ended continuation is, and whether the operators studied here transfer to it is left open (Section 10). **Judges** are Claude Opus 5 and, in the instrument section, Claude Sonnet 5 (Amazon Bedrock) and Kimi K2.6 (OpenRouter). In the loop experiments the judge is always from a different family than the generator; the idea experiment of Section 5 used Claude both to develop and to judge, a within-family judgment we flag there. Each window is judged k times ($k = 5$ in the loop experiments; $k = 3$ in the earlier ones) with independent 0–10 dimensions and the median is the score. For loop windows the dimensions are *surprise* (how unexpected the window is given the earlier text; 0 = the obvious continuation, 10 = startling yet not random), *connection* (does it bring together two distant regions of the earlier text, or an old region with something new, in a way that makes sense) and *coherence* (does it hold together as text; 0 = word salad or document boilerplate); for ideas, a nearest-equivalent plus novel-delta rubric. The full judge prompts, model identifiers, parameters and dates are in Appendix A. **Instrument calibration** (Section 7) measures the intra-window spread of the k judgments and the judge's agreement with a second judge family and with human raters. **Objective novelty**, where a public corpus exists, is computed with infini-gram [17] against the OLMo-2 corpus.

Unit of analysis and statistics. A *cell* is one premise run under one condition for 4,500 generated tokens; each condition has ten cells, one per premise, and conditions share the ten premises. The unit of inference is the cell: the windows of a cell are averaged into one score per dimension, and every comparison between two conditions is a paired comparison of ten cell means. We report the mean over cells with a bootstrap confidence interval over cells, and for each pair of conditions the mean paired difference with its bootstrap CI, an exact sign-flip permutation p-value on the ten paired differences (all 2^{10} sign patterns), and Cliff's δ on the cell means. Where a question involves several comparisons we report Benjamini–Hochberg q-values within that family alongside the raw p. Window-level statistics (bootstrap on windows, Mann–Whitney) are reported only in the appendix, as descriptive, because windows from the same stream are not independent. This seed-level analysis and the window protocol below were adopted after an external review of a first version of this manuscript, which analyzed windows as the unit; the arms of battery 3 were generated after that change, the earlier arms were re-judged and re-analyzed under it.

3.2 The anti-probable sampler

At each step the model's next-token distribution is computed as usual. Tokens below a relative probability floor ($p < 0.05\, p_{\max}$, the min-p rule) are never chosen. If the entropy of the distribution falls inside a band $[H_{\min}, H_{\max}]$ (the model is undecided but not lost), the remaining candidates (at most 128) are re-scored by

$$\text{score}(t) = \log P(t \mid \text{context}) + \lambda \cdot z\big(d(t,\ \bar{c})\big),$$

where d is the cosine distance between the token's input embedding and an exponential moving average $\bar{c}$ of the recent context in the model's own embedding space (half-life 16 tokens in the sampler experiments, 48 in the loop), and z standardizes the distances across the step's candidates so that λ reads in nats per standard deviation; the token is then sampled from the softmax of the scores. Below the band the model is confident and we let it be; above it the distribution is a genre fork and pushing there produces collapse. The band was $H_{\min} = 2.0$, $H_{\max} = 4.5$ nats in the sampler experiments and $[1.8, 4.5]$ in the loop's drift regime; the values were set on calibration probes of Qwen3-8B and transferred unchanged to the other models. Optional terms used in the loop are a *habituation* penalty (the probability of a token that occurred in the last 512 positions is divided by 1.15^m, where m is the number of its occurrences in that window, before the floor and the re-scoring) and a *bridge* bonus (weight 1.5 in the full scaffold) that rewards candidates close to anchor regions visited long ago and far from the recent context.

3.3 The reverie loop

Forced continuation without a task. A cell starts from a single sentence, the *premise* (e.g. *"The town had two clocks, and nobody remembered which one had been right first."*), and nothing else: no instruction, no role, no question. A base model continues it, and from then on its only input is its own output: each token is predicted from the premise plus everything the model has written. In all arms but one the end-of-text token is masked, so the stream never stops by itself; the model is forced to keep writing one document for 4,500 tokens. This is a deliberately hard regime, and part of what the paper measures is how a base model behaves in it.

Interruption. At chosen moments the generator is paused and a short text is *injected* into the stream as if the model had written it (pre-filled into the model's cache), and generation resumes from there. The model receives no signal that the text was injected: to it these are the next tokens of the document. Everything written before stays in the context: the model keeps all of it, and is only made to open a new sentence somewhere else. Figure 1 shows a real interruption. The default injected texts are four neutral subject changes used in rotation (*"That night she dreamed of something else entirely:"*, *"Meanwhile, in a city with no name,"*, *"There is an older story about this, and it goes:"*, *"A question nobody had asked yet:"*); other contents are studied in Section 6.

The full scaffold (DREAM). The architecture we started from adds, around the same stream: a *salience monitor* over the stream's own telemetry (a fast moving average of token embeddings and the step entropy), which fires on a semantic *jump* (the average moved far from where it was 32 tokens earlier), a *crystallization* (entropy dropped after a stretch of wandering), a *recurrence* (the average came close to a region visited long ago and not recently), and *stagnation* (it has not moved for a long stretch); a surface *genre-collapse* detector (lexical diversity, capitalization, layout symbols); a judge (Claude Sonnet 5, $k = 3$) called on salience events during generation, whose passing verdict (median score ≥ 5) triggers *escalation* (a narrower regime plus an injected textual return to the premise); a *kick* on stagnation (a stretch of harder push),

Premise: *Every morning the baker counted the loaves twice, and every morning the count was different.*

. . . the stream, 2,000 tokens later: *There was never more than thirty, but there were always twenty-one or twenty-two. That morning he counted them three times, but he got no nearer to the number. He couldn't understand it. He opened the door of the oven and pulled out a loaf of bread. He cut it open. It was all*

Injected: *Meanwhile, in a city with no name,*

The model continues: *an old woman sat at a window with her cat. She had no idea who she was or where she came from, or why she'd been left alone in this strange city with this strange cat. All she knew was that every day something went wrong with the cat: It would grow ill for no reason and then recover; it would lose its fur for days; even die for a few hours! Every day this happened, but every day it also recovered. And then one day, when the old woman looked out. . .*

Figure 1: A real interruption (Qwen3-30B-A3B, seed 2, period 300). The injected sentence is indistinguishable from the model's own text; the baker remains in the context and can be returned to.

and after repeated stagnation a subject change with *selective forgetting*: the working memory is rebuilt in a fresh cache from the premise, up to three judged-good windows and the last 200 tokens, plus the new seed, so that a deep well cannot pull the seed back; regions visited persist as anchors for the bridge term. The full scaffold is the only arm in which a judge is called inside the loop. In the main generator's scaffold cells the judge was on and reviewed 80 salience events; none reached the threshold, so escalation and the judged re-encounter never fired and the judge changed no generated token; in the replications on the other two generators the scaffold ran without the in-loop judge. In practice, then, the scaffold's active mechanisms were the kick and the reseed with forgetting, and in every arm of this paper the scores come from a judge that sees the stream only after generation. Every one of these mechanisms was born from a calibration probe in which a base model without a task fell into a specific degeneration mode (early end-of-text, erudite rumination, literal four-sentence orbits, website footers, translation tables). The generator's pretraining diet was the dominant variable in those probes: OLMo-2 sinks into footers even with forgetting; Qwen3-30B drifts in prose. With 3,000 tokens of boilerplate in the cache, every injected seed was pulled back within about twenty tokens: the accumulated context is the well, and forgetting is what lets a seed take. The results below show that this architecture is not what carries the effect; it is kept as the origin of the study and as one arm.

Conditions. Table 1 lists the arms. All loop arms use $\lambda = 0$ (no anti-probable push), since the push had no detectable effect inside the loop (Section 4); the arms differ in whether the habituation penalty is on, in whether the end-of-text token is masked, in whether the context is preserved or reset at an interruption, and in whether, when, how often and with what the stream is interrupted. Ten narrative premises (Appendix A) are shared by all arms; the sampler's random-number seed is the same (0) in every cell, so two arms that make identical decisions up to a point produce identical tokens up to that point, and the premise is the only source of variation between the cells of an arm.

Judging the stream: generated-only windows. After generation, each cell is scored on windows made only of model-generated tokens. In an interrupted cell, a window of 96 tokens starts 32 tokens after the end of each injected sentence (the *post-interruption window*); further windows start 160, 300, 450, 600 and 750 tokens after the injection when the segment is long enough, and no window crosses an injection. In a cell without injections the same window shape is cut on a uniform grid every 150 tokens. The judge sees the 600 tokens preceding the window as context (which may contain an injected sentence) and grades only the window, on the three dimensions, five times; the median is the window's score. Up to six post-interruption windows per cell, evenly spread over the stream, and three per deeper offset are judged. The primary

arm	habituation	interruption
bare	off	none: continuous generation, EOS masked
bare + habituation	on	none
habituation, EOS allowed	on	none; the model may emit end-of-text and start a new document
habituation 1.3	on (stronger)	none
interruption, no habituation	off	neutral subject change every 150 or 300 tokens, context preserved
habituation + interruption 150	on	neutral subject change every 150 tokens, context preserved
salience only	on	none; salience events mark review windows
DREAM scaffold	on	full scaffold: salience, in-loop judge, kick, reseed with forgetting, re-encounter
content: re-encounter / premise / own past	on	every 150 tokens: a return-to-the-premise stitch / the premise itself / a window of the stream's own past (≥400 tokens back)
salience: re-encounter	on	the stitch injected on each salience event, no judge gate
period 75 / 300 / 600 / 900	on	neutral subject change at other periods (900 also with the stitch)
sham break 300	on	a paragraph break ("\n\n") every 300 tokens, nothing else
sham continuity 300	on	"And so, as before," every 300 tokens: a boundary that asks for continuity
reset + subject change 300	on	neutral subject change every 300 tokens on a *reset* context (premise + injected sentence only)
reset + break 300	on	a paragraph break every 300 tokens on a reset context

Table 1: Conditions: the habituation × interruption factorial with its baselines (top), the content and timing arms (middle) and the boundary and context controls (bottom). Ten premises each, 4,500 tokens per cell, main generator Qwen3-30B-A3B-Base; the ladder bare / habituation / habituation + interruption 150 / scaffold is replicated on Qwen3-8B-Base and OLMo-2-13B.

measure of a cell is the mean of its post-interruption windows (grid windows in uninterrupted arms); the deeper offsets give the decay curve of Section 6. Each cell stores its token stream and the exact position of every event and injection. The protocol of the first version of this study, which cut 160-token windows right before each injection and on a 150-token grid so that a window could contain an injected sentence, is kept in Appendix C for comparison.

4 Two motivating nulls: the sampler and the prompt

The program began where effort is usually spent, the sampling operator and the input, and found nothing at the level of ideas in either place. Both studies are small and are reported here as motivation for the loop experiments, with their limits stated; the full tables are in the repository.

The sampler doubles surface novelty. On OLMo-2-13B, whose training corpus is public, we ran a fully paired grid of 3 prompts × 5 seeds × {min-p baseline, $\lambda = 1$, $\lambda = 2$} and counted, with infini-gram, which n-grams of each generation occur anywhere in the corpus (Figure 2). Novel 4-grams rise from 21.5% (baseline) to 36.0% ($\lambda = 1$) and 45.5% ($\lambda = 2$): paired bootstrap Δ +14.6 pp $[7.9, 21.9]$ and +24.0 pp $[18.2, 29.5]$, monotone in λ and present

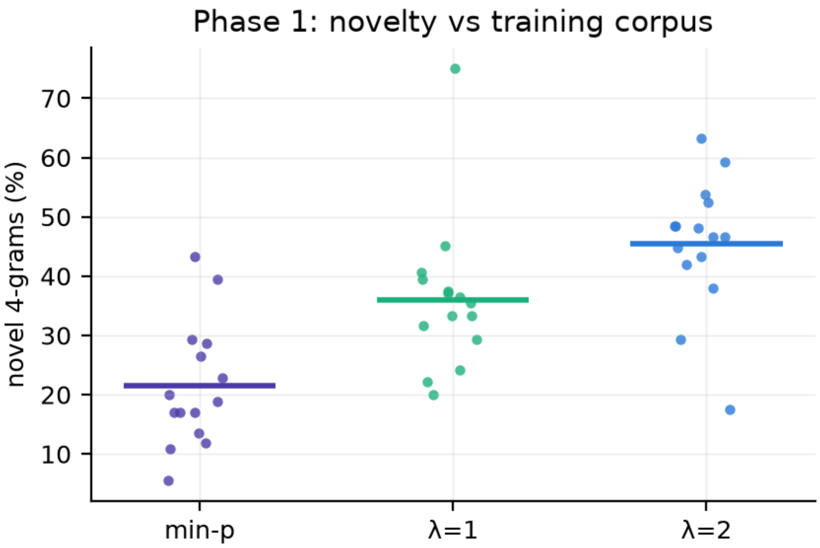

Figure 2: Objective novelty against the OLMo-2 training corpus: novel 4-grams per cell, 3 prompts × 5 seeds. The anti-probable push roughly doubles surface novelty over min-p sampling.

in each prompt separately. The share of generations containing a verbatim training block of eight or more words falls from 0.80 to 0.20 in both machine arms. The coherence cost is about +1.2 perplexity under a cross-family model, uncorrelated with novelty within arms (Spearman +0.12). Three escape modes were mapped (recitation, collage and factual paraphrase), and the entropy ceiling of the band, above which the model is at a genre fork and pushing produces collapse, transferred across model families.

And has no detectable effect at the level of ideas. Two tests. In verified search (online bin packing, the FunSearch benchmark, with a deterministic verifier in place of a judge; evolutionary loops of 5 runs × 8 generations × 20–40 candidates on Qwen3-8B and Qwen3-30B-A3B, 3,200 verified heuristics on the larger model) the anti-probable operator does not separate from plain sampling: both arms rediscover best-fit at will and neither beats it on held-out instances. And inside the reverie loop of Section 6, the same push against plain sampling in the same scaffold shows no detectable difference in judged surprise, connection or novel delta across 1,335 window-level judgments by two judges under three rubrics (the surprise rubric: 2.97 vs 3.48, CI of the difference [-1.50, +0.49]). With ten cells per arm and a judge whose per-window noise is about ±0.7, an effect of half a point would not have been detected; the conclusion is a null at the present resolution, not an absence.

5 The prompt

We composed three kinds of input (a typical request, a distant concept pair drawn from a band of embedding distances, and a composed improbable context) plus two ablations (fragments only, register only), had Claude Opus 5 develop each into an idea, and had Claude Sonnet 5 judge the result against its nearest known equivalent ($k = 3$; $n \approx 15$ per arm; developer and judge from the same family, the one within-family judgment in this paper). Input improbability was measured two ways: semantic kNN distance to 10,000 real instructions (Alpaca, embedded with all-mpnet-base-v2) and perplexity. The typical prompt matched or beat every improbable arm, two of which scored lower by a margin whose interval excluded zero; improbability did not correlate with judged novelty ($|r| < 0.2$, Figure 3); a pilot effect ($n = 8$) had not replicated.

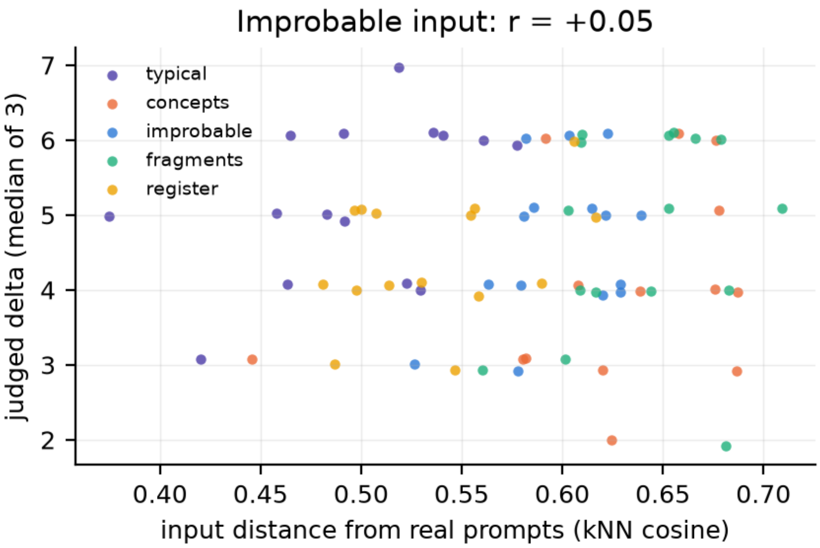

Figure 3: Judged novel delta of the developed idea against the input's semantic distance from 10,000 real prompts, per arm. No relationship.

We found no benefit from prompt improbability under the tested operationalizations; whether other operationalizations, developers or judges would find one is open.

6 The loop

All results in this section use Qwen3-30B-A3B-Base unless stated, ten premises, 4,500 tokens per cell, $\lambda = 0$, Claude Opus 5 as judge with $k = 5$, generated-only windows, and the cell as unit: numbers are means over cells (one per premise), intervals are 95% bootstrap CIs over cells, and paired differences come with an exact sign-flip permutation p on the ten paired cells and Cliff's δ on cell means. Full tables, including the Benjamini–Hochberg q-values within each family of comparisons and the event-protocol tables of the first version, are in Appendix D. Every figure shows one point per premise.

6.1 Forced continuation degenerates; the scaffold revives it

Left to continue a premise with nothing else, the base model degenerates within a few hundred tokens into a repetitive or corpus-like mode and stays there: *"Highly recommended. Highly recommended. . . . "* for two thousand tokens; a table of three-digit numbers; an English-exam answer key. Judged surprise 0.45 $[0.28, 0.68]$, connection 0.30, coherence 2.52 (ten cells). The full scaffold brings the stream to 2.70 / 1.85 / 6.02, and every premise moves in the same direction (Figure 4). But the scaffold is not the smallest thing that does this.

6.2 Taking the loop apart: habituation and interruption

The ladder. Two operations of the scaffold suffice, and they are not the elaborate ones (Figure 4, Table 2). Adding only *habituation* to bare generation (a windowed repetition penalty, no interruption) removes the literal orbits: surprise 0.45 → 1.58 (Δ +1.13 $[+0.72, +1.47]$, $p = 0.004$, $\delta = +0.80$), connection 0.30 → 1.28, coherence 2.52 → 4.45; the stream is fluent and, to the judge, still not very surprising. Adding a neutral subject change every 150 tokens over the preserved context takes it to 3.02 / 3.68 / 6.12: over habituation alone, surprise Δ +1.43

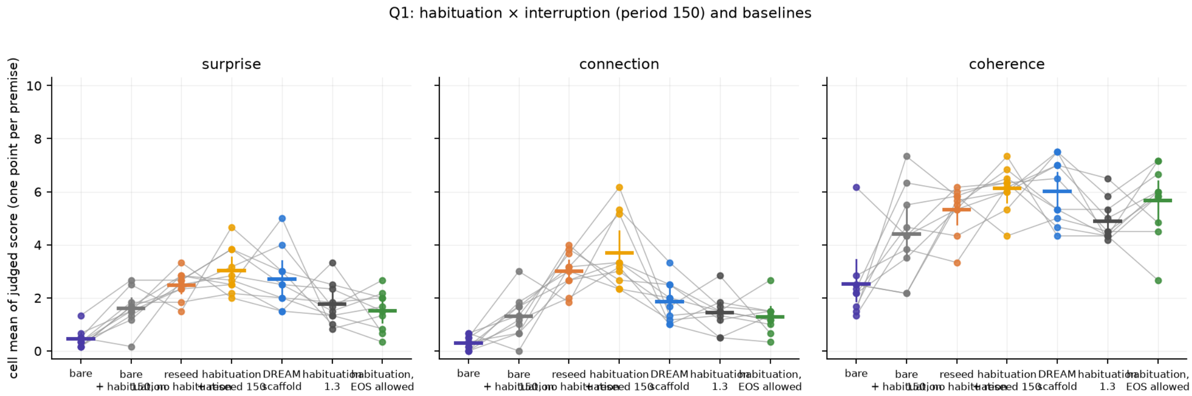

Figure 4: Habituation and interruption: bare, bare + habituation, interruption without habituation, habituation + interruption 150, the full scaffold, and the two habituation baselines (stronger penalty; EOS allowed). One point per premise, grey lines join the same premise across arms, bars are cell means with 95% bootstrap CIs over cells (Qwen3-30B-A3B, generated-only windows).

$[+0.93, +2.05]$ ($p = 0.002$, $\delta = +0.85$), connection $+2.40$ $[+1.58, +3.32]$ ($p = 0.002$, $\delta = +0.93$), coherence $+1.67$ $[+0.27, +2.88]$ ($p = 0.05$). This plain interruption matches the full scaffold on surprise (3.02 vs 2.70) and beats it on connection (3.68 vs 1.85; the scaffold's gain over habituation is $+0.57$ $[+0.17, +1.02]$) at equal coherence (6.12 vs 6.02): the salience monitor, the in-loop judge (which never fired) and the forgetting reseed add nothing detectable over the plain interruption, and the scaffold gives back connection. Battery 3 will show that this is not the forgetting itself (a complete reset every 300 tokens scores at least as well as the preserved context) but, most likely, how rarely the scaffold interrupts: 0–3 reseeds per cell, after three consecutive stagnations, against 30 for the clock. Habituation accounts for part of the surprise; the interruption accounts for the rest and, on these all-windows numbers, for most of the additional connection beyond habituation; the self-copy check below will cut the connection number down and leave the surprise one.

The factorial and the baselines. Battery 3 completes the 2×2 (Table 3, 9). The neutral subject change *without* habituation reaches 2.45 / 3.02 / 5.38 at period 150 and 2.49 / 1.97 / 5.50 at period 300: above habituation alone on surprise ($+0.87$ $[+0.27, +1.43]$, $p = 0.04$; $+0.91$ $[+0.43, +1.40]$, $p = 0.016$) and, at 150, on connection ($+1.73$, $p = 0.010$), and below the same interruption with habituation by 0.4–0.6 on surprise and connection and 0.7–0.9 on coherence, differences whose intervals include zero at $n = 10$. The two operations are closer to additive than to interacting, and the interruption is the larger of the two: alone it gets most of the way, and habituation adds a fluent stream between injections. Two baselines say what habituation is not. A stronger penalty (1.3 instead of 1.15) changes nothing (1.77 / 1.43 / 4.88; Δ $+0.18$ $[-0.22, +0.65]$ on surprise). Allowing the end-of-text token, so that the model may close its document and open another (a boundary of its own choosing), gives more coherent text (5.69, $+1.24$ $[+0.10, +2.24]$) and no more surprise or connection (1.59 / 1.33; $+0.01$ and $+0.04$): the model's own document breaks are boundaries without a subject, and they do what a sham break does below.

Self-copy: what the window judge cannot see. Before going further, a check that the document-level judgment (Section 6.5) forced on us. A base model that is given the same four subject-change sentences in rotation learns the rotation: after *"That night she dreamed of something else entirely:"* it tends to write what it wrote the last time that sentence appeared, 600 or 1,200 tokens earlier. We flag a judged window as *copied* when at least half of its 12-token shingles occur earlier in the same stream, and note whether the source lies within the 600

arm	Qwen3-30B-A3B			Qwen3-8B			OLMo-2-13B		
	S	C	H	S	C	H	S	C	H
bare	0.45	0.30	2.52	0.43	0.40	3.88	1.28	0.93	3.03
bare + habituation	1.58	1.28	4.45	1.37	1.00	4.35	1.96	1.26	3.99
habituation + interruption 150	**3.02**	**3.68**	**6.12**	**2.76**	**3.17**	5.15	2.77	**2.87**	4.05
DREAM scaffold	2.70	1.85	6.02	2.70	2.35	**5.38**	**3.60**	2.23	**6.23**

Table 2: The ladder on three generator models from two families: judged surprise (S), connection (C) and coherence (H), means over cells (generated-only windows, Opus 5, $k = 5$; ten premises per arm). Habituation buys part of the surprise; the interruption buys the rest and most of the additional connection; the scaffold adds nothing detectable over the plain interruption on the Qwen models and, on OLMo, adds surprise and coherence (its selective forgetting lifts the stream out of a web-boilerplate well) while giving back connection.

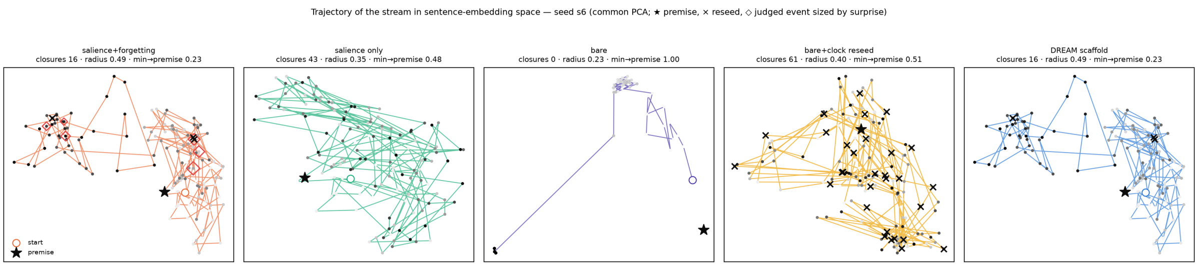

Figure 5: Where the stream goes: trajectories in sentence-embedding space (64-token windows, one PCA per seed), seed 6. ⋆ premise, ∘ start, × reseed, ⋄ judged event sized by surprise. Bare makes one jump and freezes; the scaffold orbits and returns; the clock reseed jumps across the space keeping its context.

tokens the judge sees (Table 22). Bare generation is 67% copied (its loops, all visible to the judge, who scores them 0.03); habituation alone 27%; the clock arms with the fixed rotation 65–80% at periods 150 and 300 (at 300, the source of 62% of all windows lies outside the judge's context), 38% at 600 and 12% at 900; the reset arms 0%; the scaffold 6%. A copied window scores high on "connection" for the obvious reason (it *is* earlier text) and, when its source is out of view, on surprise too. So we recompute the ladder on *fresh* windows only. The interruption still lifts the stream over habituation: at period 150 surprise +1.20 $[+0.43, +2.11]$ ($p = 0.010$), connection +0.75 $[+0.38, +1.17]$ ($p = 0.008$), coherence +1.72; at 300 +1.32 $[+0.73, +1.88]$ ($p = 0.004$), +0.75, +1.45; at 900 +1.35, +0.99, +1.67; and the reset arm, which cannot copy, +1.93 $[+1.59, +2.28]$, +2.03 $[+1.75, +2.33]$, +1.82 (all $p \leq 0.004$, every premise). The surprise effect of the interruption is therefore real on generated, fresh text and about one point smaller than the all-windows number; the connection effect is largely a self-copy artefact (+2.40 on all windows, +0.75 on fresh ones), and the reset arm's advantage over the preserved context on connection (+1.28 $[+0.82, +1.73]$ on fresh windows) is in good part that it does not replay itself. From here on we report both numbers where they differ, and we treat the fresh-only ones as the primary estimate of what the interruption does. The lesson for the recipe is plain: a fixed rotation of four sentences is a cycle the model will close; give it a new sentence each time, or reset.

Where the stream goes. In sentence-embedding space (Figure 5) bare generation has explored radius 0.18 and mean step 0.14 and ends 0.78 from the premise; the scaffold has radius 0.51 and returns to within 0.57 of the premise; the clock reseed spreads over the space in jumps while keeping the whole context; the reset arm of battery 3 (Section 6.3) will show that keeping it is not what the judge rewards.

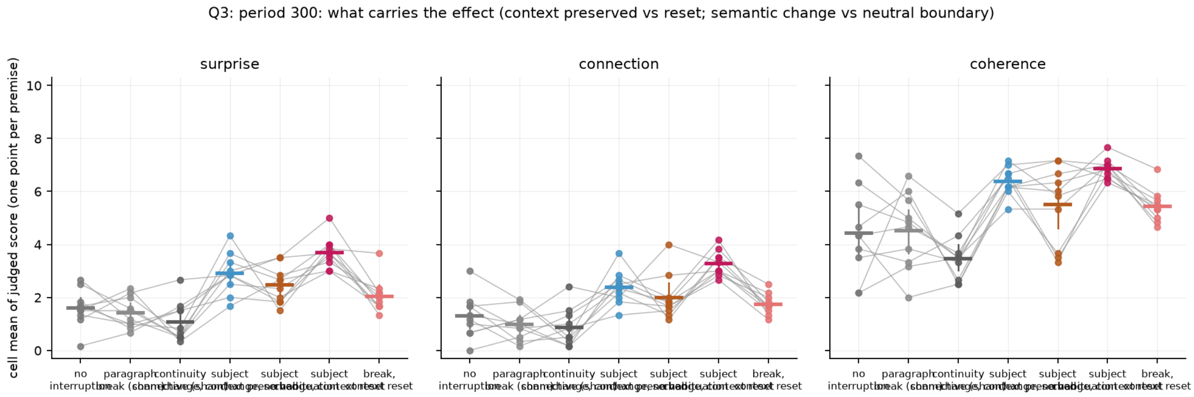

Figure 6: What carries the effect at period 300: no interruption, a paragraph break (sham), a continuity connective (sham), the subject change on the preserved context, the subject change without habituation, the subject change on a reset context, and a break on a reset context; one point per premise.

6.3 What carries the effect: boundary, content, context

An interruption is three things at once: a boundary in the text, a new subject, and a context that is kept or dropped. Battery 3 separates them at period 300 (Figure 6, Tables 7 and 9). A bare paragraph break every 300 tokens (a sham interruption) leaves the stream where habituation alone leaves it (1.42 / 0.97 / 4.43 vs 1.58 / 1.28 / 4.45). A connective that asks for continuity (*"And so, as before,"*) is, if anything, worse than nothing (1.08 / 0.86 / 3.49; -0.50 [-1.20, +0.25] on surprise, -0.96 on coherence): a boundary that points back is what the premise and own-past injections were. The neutral subject change over the preserved context gives 2.90 / 2.38 / 6.40 ($+1.32$ $[+0.68, +1.93]$ on surprise, $p = 0.004$; $+1.10$ on connection, $p = 0.014$; $+1.95$ on coherence, $p = 0.02$; against the sham break $+1.48$ / $+1.42$ / $+1.97$, all $p \leq 0.006$). And the same subject change on a *reset* context, where the model keeps only the premise and the injected sentence, gives 3.72 / 3.32 / 6.88: higher than the preserved context on every dimension ($+0.82$ $[+0.32, +1.30]$, $p = 0.02$; $+0.93$ $[+0.53, +1.32]$, $p = 0.004$; $+0.48$ $[+0.03, +0.95]$), and every premise moves the same way on surprise. Resetting without a new subject (a break on a reset context) gives 2.03 / 1.73 / 5.43, little more than the sham; the subject change against the break is $+1.68$ / $+1.58$ / $+1.45$ under reset and $+1.48$ / $+1.42$ / $+1.97$ under the preserved context ($p \leq 0.006$). What carries the effect, then, is the new subject; the boundary by itself does nothing; and dropping the earlier text from the model's context does not hurt what the judge reads, it helps. Two things must be said about that last result. Under reset the model cannot integrate the earlier text, and its "connection" is what it can still do: return to the premise, sometimes verbatim, in a fresh mini-story that the judge, seeing 600 tokens of earlier text about the same premise, scores as connected and coherent; under the preserved context the model can integrate, returns less, and is entrained by 3,000 tokens of its own past. Connection as this rubric measures it does not separate a long-range integration from a return to the shared beginning; the preserved context is a choice of the recipe we studied, not a demonstrated ingredient of the judged effect, and Section 6.5 asks the question at the level of the whole document, where the difference between integration and restart should show if it exists.

6.4 A confirmatory replication on new premises

Everything above was designed and analyzed in sequence, and the questions of batteries 2 and 3 were formulated after seeing battery 1. Before any result of battery 3 or of the generated-only judgments was seen, we wrote ten new premises (Appendix A), fixed a different random seed, chose five arms at period 300 (habituation alone, the sham break, the subject change on the preserved context, the subject change without habituation, and the subject change on a reset context) and pre-registered four contrasts in the laboratory notebook: H1 (primary), the subject change on the preserved context beats habituation alone on surprise; H2, it beats the sham break; H3, it beats the reset context on connection; H4 (two-sided), habituation matters given the interruption. Fifty new cells were then generated and judged under the same protocol (Table 27, one-sided exact permutation tests). H1 is confirmed: $+1.48$ $[+0.72, +2.05]$, $p = 0.003$ (on fresh windows only, $+0.82$ $[+0.03, +1.45]$, $p = 0.04$). H2 is confirmed on all windows ($+0.91$ $[+0.23, +1.55]$, $p = 0.02$) and not on fresh ones ($+0.71$, $p = 0.11$). H3 is refuted in the direction battery 3 had found: the reset context scores *higher* on connection (-0.55 [-1.03, -0.03] for preserved minus reset; -1.02 on fresh windows), and on surprise (4.17 vs 2.87). H4 is not supported ($+0.51$ [-0.02, +1.03], $p = 0.12$ two-sided). The document-level judgment repeats on the new premises as well: the reset arm above the preserved context on all four document dimensions ($p \leq 0.03$), the preserved context at or below habituation alone (Table 30). The replication thus confirms the effect the paper rests on, on premises and a random seed the program had never seen, and it settles the two questions the exploratory batteries had left open in the less flattering direction: the context need not be kept, and habituation is not what makes the interruption work.

6.5 The whole document

Every score above is local: a window of 96 tokens read against 600. Does the interrupted loop build anything over 4,500 tokens? We had the judge read the *whole* stream of each cell, with the injected sentences removed, and score four things once (Opus 5, $k = 3$, median; one score per cell): *integration* (are characters, motifs or ideas from earlier parts taken up later and joined), *development* (does something build rather than restart or repeat), *coherence* (does it read as one text) and *surprise* (does the whole go somewhere unpredictable yet sensible). The answer is no, for every arm (Figure 7, Table 25): no condition exceeds 2.5 on any document dimension. Worse for the recipe: judged as a whole, the interrupted stream over the preserved context is a sequence of restarts and reads *below* the uninterrupted habituated stream (period 150: surprise 0.10 vs 1.40, -1.30 [-1.60, -1.00], $p = 0.002$; development 0.30 vs 1.10; period 300: surprise -0.90, $p = 0.03$; integration -0.60; the scaffold is at the level of habituation). The judge's notes say why: the interrupted stream "repeats the same four-part block verbatim" and "cycles four short fragments", the self-copy of the paragraph above seen whole. The longer periods and the stitch arms, with fewer replays and longer stretches, sit between (600 and 900: 1.2–1.8 on every dimension). The reset arm reads best of all (integration 2.00, development 1.60, coherence 2.20, surprise 2.50; above the preserved context on all four, $p \leq 0.03$, and above habituation on surprise, $+1.10$, $p = 0.008$), because each of its parts is a clean piece that returns to the premise, which the document judge, like the window judge, credits as a kind of unity. So the local gains of Sections 6.2 to 6.8 do not compose: an interruption is a local operation that makes the next hundred tokens more surprising, connected and coherent than they would have been, and leaves the document a collection. Keeping the earlier text in the model's context, which is what a writer would want, produces no integration that either judge can see; the accumulation the architecture was built for does not happen. We take this as the clearest statement of what the paper has and has not found: an operator on windows, not on wholes.

6.6 The interruption must lead away

With the same period (150), a neutral subject change and a re-encounter stitch that asks the stream to return to its beginning are indistinguishable (3.02 / 3.68 / 6.12 vs 2.83 / 3.70 / 5.50; every paired CI covers zero). Injecting the *premise itself* (1.16 / 1.11 / 3.91) or a 64-token

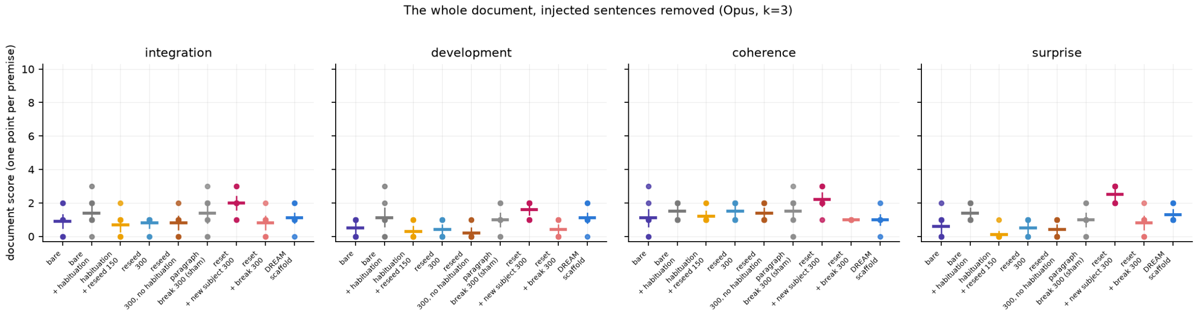

Figure 7: The whole document, injected sentences removed: integration, development, coherence and surprise of the 4,500-token stream (Opus 5, $k = 3$; one point per premise). No arm builds a whole; the interrupted stream over the preserved context reads as a sequence of restarts; the reset arm's clean parts, each returning to the premise, read best.

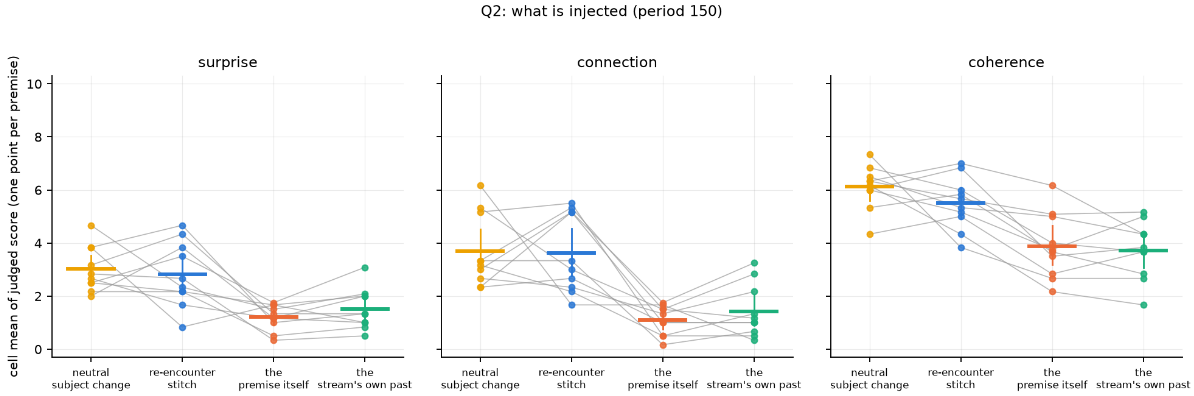

Figure 8: Content of the interruption at period 150: neutral change, re-encounter stitch, the premise itself, the stream's own past. Injecting a return is as bad as not interrupting.

window of the stream's *own past* (1.48 / 1.44 / 3.70) is as bad as not interrupting at all: against the neutral change, surprise -1.86 [-2.38, -1.35] and -1.54 [-2.17, -0.86], connection -2.58 and -2.24, coherence -2.21 and -2.42, all $p \leq 0.006$, δ -0.84 to -1.00 (Figure 8). Qualitatively, the own-past injection reinforces whatever mode the stream is in: an exam-key well was fed its own questions back. A return works only as a short stitch that still opens a new sentence; a literal return closes the loop.

6.7 Timing: salience against the clock

Salience events (jump, crystallization, recurrence; 1–9 per cell, median ≈5) are the moments the scaffold's monitor picks out of the stream's own telemetry. As a policy for *which* windows to judge they beat a clock in the first version of this study; the question here is whether they are also the right moments to *interrupt*. With the same stitch, timing by salience events and timing by a clock of matched frequency (every 900 tokens) are indistinguishable on generated text: the post-interruption windows score 3.04 / 2.92 / 5.75 against 3.05 / 2.71 / 5.72 (paired differences -0.01 [-0.76, +0.73] on surprise, +0.21 on connection), and the windows at least 300 tokens after the last injection, where the stream is left alone, 1.82 / 1.80 / 4.20 against 2.08 / 1.71 / 4.73 (-0.26 [-0.78, +0.21]) (Figure 9, Table 12). Salience monitoring by itself, with nothing injected, leaves the stream where habituation alone leaves it (1.65 / 1.30 / 4.63 vs 1.55 / 1.28 / 4.48). The first version of this study reported a large stream-level advantage of the clock over salience timing (4.7 vs 2.2 on uniform windows); that advantage was carried by the injected sentences inside the clock arm's windows, of which it has more. What remains is a null

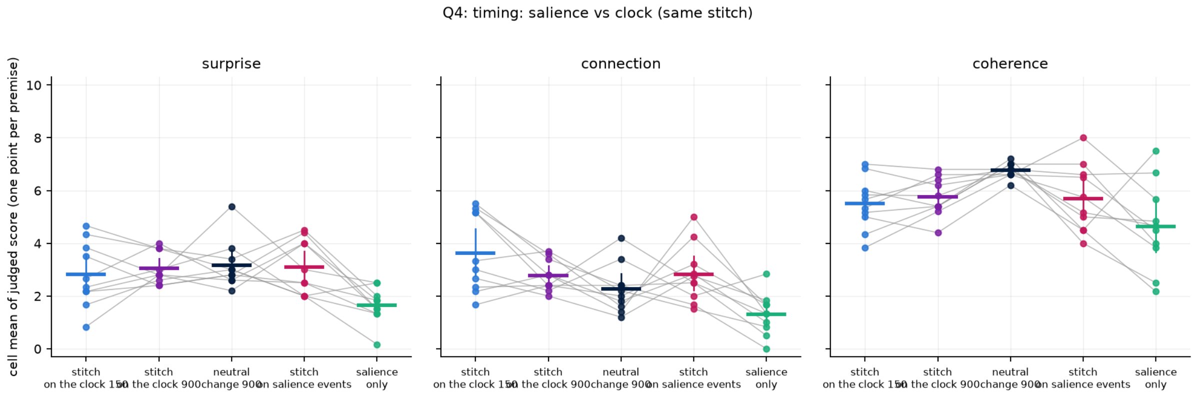

Figure 9: When to interrupt: the re-encounter stitch on the clock (150 and 900), the neutral change every 900, the stitch on salience events, and salience only (nothing injected). Post-interruption windows; one point per premise.

at the present resolution: salience is a good reader of the stream and, as a metronome, neither better nor worse than a clock ticking at the same rate. Since a clock is free and regular, the clock is what we recommend.

6.8 Period and decay

The first version of this study reported that the yield of an interruption grew with the length of the thread it broke (5.3–5.9 after 300–900-token threads against 3.1 after 150) and decayed over the next few hundred tokens, and named 300 tokens as the best rhythm. With the injected sentence excluded from the judged window, that finding largely disappears (Figure 10). The post-interruption window (32–128 generated tokens after the injection) scores 3.02 at period 150, 2.87 at 300, 2.88 at 600 and 3.18 at 900 on surprise, every paired difference against 150 covering zero; connection is highest at the shortest period on all windows (3.68 vs 2.38, 2.02 and 2.22; Δ -1.30 to -1.67, $p \leq 0.008$), a difference that is self-copy (Table 22: on fresh windows connection is 2.0 to 2.3 at every period), and coherence rises a little with the period (6.12 $\rightarrow$ 6.43, 6.25, 6.78). Deeper into a segment the surprise of generated text holds for a few hundred tokens and then sinks (period 900: 3.18, 2.67, 3.17, 2.77, 2.23 and 1.80 at 32, 160, 300, 450, 600 and 750 tokens after the injection, Figure 11), toward the level of habituation alone (1.55). The stream estimate (offset means weighted by the stretch of the segment they represent) is therefore 3.02 / 3.68 at 150, 2.76 / 2.30 at 300, 2.84 / 2.12 at 600 and 2.64 / 2.03 at 900 (Table 16). Among the tested periods, then, no period is better than another on fresh generated text, and what the longer periods buy is a little coherence and fewer replays. A period of 75 cannot be measured under this protocol (no 96-token window of generated text fits between two injections); under the event protocol its stream was dead (0.88), which is the one part of the earlier claim that survives: a thread needs some tens of tokens to exist before it can be broken. The larger numbers of the first version were, in good part, the judge reading the injected sentence. What survives is the direction of every effect and an operating range: a break every 150–300 tokens.

6.9 Three generator models from two families

The ladder replicates on Qwen3-8B-Base and OLMo-2-13B (Table 2, Figure 12). On the 8B: 0.43 $\rightarrow$ 1.37 $\rightarrow$ 2.76 (interruption over habituation: surprise +1.39 $[+0.72, +2.07]$, $p = 0.008$; connection +2.17 $[+1.53, +2.83]$, $p = 0.002$), scaffold 2.70 / 2.35. On OLMo, whose bare stream wanders across web genres rather than locking into literal orbits and is therefore less dead (1.28),

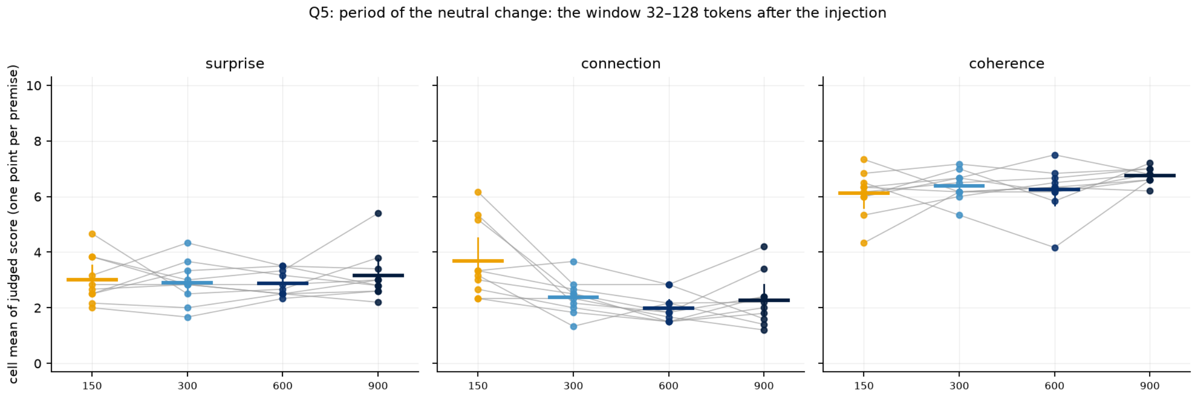

Figure 10: Period of the neutral change: the post-interruption window (32–128 generated tokens after the injection) at periods 150, 300, 600 and 900; one point per premise.

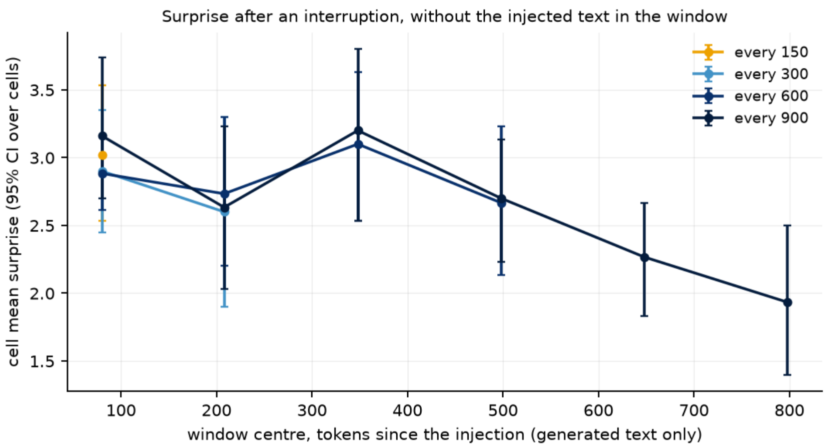

Figure 11: Decay: cell-mean surprise of generated-only windows by tokens since the injection, per period.

the interruption over habituation gains connection (+1.61 [+0.83, +2.35], $p = 0.004$) and only some surprise (+0.82 [-0.21, +1.81], $p = 0.17$), and costs nothing detectable in coherence. Self-copy is present there too (78% of the interruption arm's windows on the 8B, 45% on OLMo, Table 23); on fresh windows the interruption adds +0.95 (8B) and +1.0 (OLMo) surprise over habituation. On OLMo the full scaffold is the best arm on surprise (3.60, +1.64 over habituation, $p = 0.04$) and coherence (6.23, +2.24, $p = 0.02$), with less connection than the plain interruption (2.23 vs 2.87). What the scaffold does for OLMo is what a reset does: pull the stream out of a web-boilerplate well, the mechanism the calibration probes had suggested; on a generator whose well is boilerplate, the reset earns its keep, and battery 3 says it costs nothing on the one that drifts in prose. Quantization is not what carries any of this: the same ladder on the unquantized (bf16) Qwen3-8B-Base gives $0.60 \to 1.33 \to 2.70$ on surprise and $0.47 \to 1.06 \to 3.25$ on connection (interruption over habituation +1.37 [+0.26, +2.42] and +2.19 [+1.27, +3.10]), against $0.43 \to 1.37 \to 2.76$ and $0.40 \to 1.00 \to 3.17$ at 8 bits.

A post-trained model. As one step outside the scope decision, we ran the same ladder on the post-trained Qwen3-8B (the released chat model, 8-bit), continued raw, without a chat template, from the same premises. Its degeneration modes are its own: multiple-choice exam items with answer keys, and a chain-of-thought voice that treats the premise as a riddle (*"Wait, let me think of a classic riddle..."*). The ladder holds at a lower level: $0.60 \to 0.77 \to 1.90$ on surprise and $0.42 \to 0.75 \to 2.28$ on connection; habituation does less for it than for the base

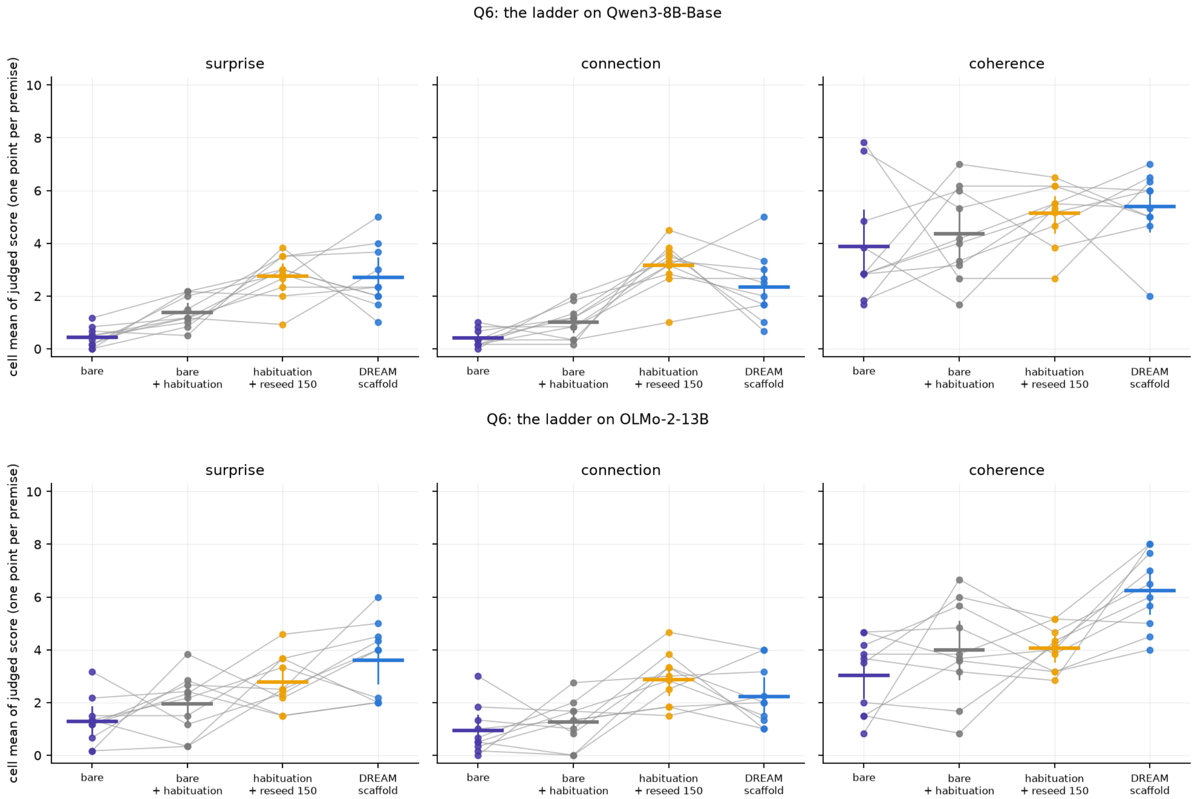

Figure 12: The ladder on Qwen3-8B-Base (top) and OLMo-2-13B (bottom); one point per premise.

models (+0.17 on surprise) and the interruption still adds +1.13 $[+0.58, +1.78]$ ($p = 0.002$) and +1.53 $[+0.87, +2.32]$ ($p = 0.004$). Whether the operators help such a model when it is used as intended, in a dialogue, is not tested here.

7 The instrument, measured

Every idea-level claim in this paper rests on an LLM judge, so we measured the judge before using it, in three ways: its repeatability, its agreement with a second judge family, and its agreement with human readers. Repeatability is not validity: five calls to the same model measure the stability of one reader, not whether that reader is right, and only the last two checks bear on validity.

Repeatability. On the same 89 windows judged $k = 5$ times by two models, Claude Opus 5 gives continuous scores with a median intra-window spread of 0.71; Claude Sonnet 5 has spread 0.27, but by scoring zero on the connection dimension almost everywhere; a ruler that only reads "0" is consistent (Figure 13). The three-dimension surprise/connection/coherence rubric has spreads of 0.45–0.70 under Opus. A test–retest of the whole pipeline came for free: two arms of the ablation battery (the full scaffold and its no-re-encounter ablation) turned out to be byte-identical in all ten cells, because the in-loop judge never passed and the re-encounter never fired, and they were judged independently. On their 79 generated-only windows the medians of five agree exactly in 91% of windows on surprise (mean absolute difference 0.09), 85% on connection (0.15) and 71% on coherence (0.29); the cell means differ by -0.01 [-0.09, +0.07] on surprise. Two consequences shape the paper. A binary threshold on a noisy judge is a coin (the same window scored 5.24 one night and 4.48 the next), so every result uses continuous medians of k samples. And a per-judgment noise of about ± 0.7 is one of several sources of variance, not

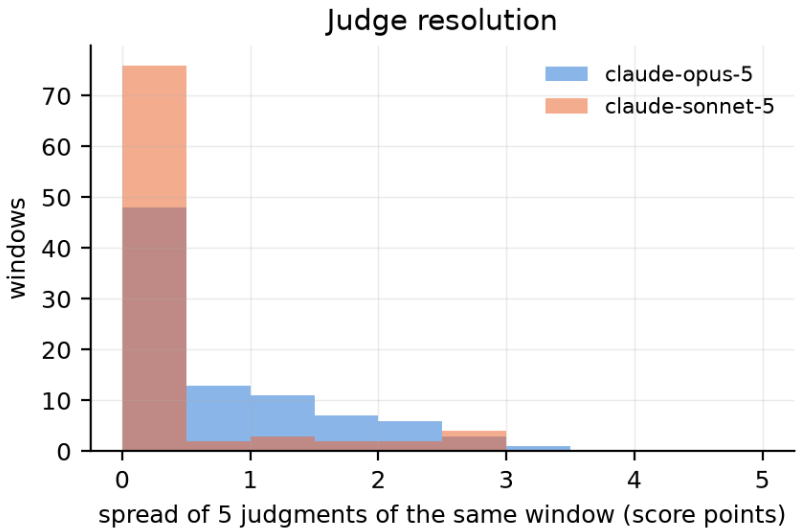

Figure 13: Spread of five judgments of the same window (delta rubric, 89 windows). Sonnet 5 concentrates at zero spread by scoring zero; Opus 5 gives continuous scores at ± 0.7 noise.

a threshold of detectability: the median of five reduces it, windows within a stream vary more than that, cells vary more still, and the power of a comparison is set by the ten paired cells, not by the judge's spread. That is why the sampler's null inside the loop (Section 4), obtained on windows and with a half-point difference, is reported as "no detectable effect at the present resolution", not "absent".

A second judge family. To check that the loop results are not the taste of one model family, a stratified sample of 140 already-judged event-protocol windows (20 per condition across bare, habituation, clock reseed, scaffold, re-encounter stitch, period 300 and salience-timed re-encounter) was re-judged by Kimi K2.6 (Moonshot, via OpenRouter) with the same rubric and $k = 5$. Agreement with Opus 5 on the median of five: surprise Spearman $\rho = +0.85$, connection $+0.77$, coherence $+0.71$ ($n = 140$ windows from 70 cells; the correlations are computed on windows and their p-values are not corrected for the clustering by cell). Kimi is more generous (means about one point higher) and noisier (intra-window spread 2.0–2.6), but every condition ordering replicates: bare 0.55 $\to$ habituation 3.15 $\to$ clock reseed 4.30 $\approx$ scaffold 4.10 $\to$ period 300 5.65 on surprise, and the neutral reseed and the re-encounter stitch highest on connection under both judges. Repeated on the generated-only protocol with a random (not stratified) sample of 160 post-interruption windows from eight conditions including the battery-3 arms, the agreement is $\rho = +0.75$ on surprise, $+0.80$ on connection and $+0.65$ on coherence, and the ordering is again reproduced (bare 1.0 $<$ habituation 2.5 $\approx$ sham break 2.1 $<$ interruption 150 3.3, 300 5.4, scaffold 4.9 $<$ reset 5.7 on surprise under Kimi). Two families agreeing does not make the rubric valid; it rules out one family's taste as the whole explanation.

Human rating, round 1. A blind rating of 63 event-protocol windows (nine per condition, stratified by judged surprise, shuffled, no labels, the earlier text shown) by three independent raters recruited on a freelance platform without knowledge of the hypotheses or conditions (Appendix B): a native English reader with professional editing experience, a native or bilingual English speaker, and an advanced non-native reader with graduate training in language. They rated surprise and coherence (not connection) with the judge's rubric. All three returned complete ratings using the whole of both scales, with distinct personal calibrations (mean surprise 3.4, 4.3 and 6.1). Each rater alone agrees with Opus 5 on surprise (Spearman $\rho = +0.40$, $+0.50$,

$+0.47$; all $p \leq 10^{-3}$) and on coherence ($+0.60$, $+0.47$, $+0.23$); the mean of the three agrees more strongly than any one (surprise $\rho = +0.58$, $p = 7 \times 10^{-7}$; coherence $+0.52$, $p = 10^{-5}$), about as strongly as the raters agree with each other (pairwise Spearman on surprise 0.71, 0.38, 0.25; Krippendorff's interval $\alpha = 0.30$; on coherence 0.69, 0.60, 0.42; $\alpha = 0.38$; the low α reflects the different scale calibrations that rank correlation ignores). The human consensus reproduces the condition ordering that carries the paper: period-300 interruption 6.5 $\approx$ neutral clock reseed 6.4 $>$ re-encounter stitch 5.3 $>$ salience-timed 5.0 $>$ scaffold 3.6 $\approx$ habituation-only 3.6 $>$ bare 1.5 (surprise, means over nine windows); interruption vs bare $[+3.1, +6.5]$, habituation vs bare $[+0.1, +3.9]$, and the plain interruption above the full scaffold by a wider margin than under Opus. On coherence the human readers rank habituation without interruption highest (7.7) and the interrupted arms 5.7–6.3: to a human reader, the subject change inside a window costs some fluency that the LLM judge does not penalize. Three limits of this round are plain. The sample was stratified by the judge's own surprise score, which widens the range and can inflate a rank correlation; connection, the dimension on which the interruption's advantage is largest, was not rated; and inter-rater agreement is low to moderate, so the human data support the ordering of conditions and a moderate agreement with the judge, not the judge's exact numbers. [round 2: 56 generated-only, fresh windows from eight conditions, one per randomly chosen cell (not judge-stratified), three dimensions including connection, five raters; results to be inserted here]

8 Inside the network: a descriptive look

Judges and sentence embeddings see the text; the model's own residual stream sees the computation. This section is exploratory and descriptive: it reports correlational geometry of the residual stream with judged windows as observations (clustered by cell; for the pooled correlations we give cluster-robust intervals from a bootstrap over cells, and the within-condition values are descriptive), uses a logit lens for one coarse quantity, and makes no causal intervention. It characterizes what the judged dimensions co-vary with in the model's state; it does not establish a mechanism. We re-run every finished stream through its generator (one forward pass with a KV cache, exact token positions) and capture the residual at 13 layers (every 4th of 48 for Qwen3-30B-A3B; every 3rd for the 36-layer Qwen3-8B and the 40-layer OLMo-2), mean-pooled over 64-token windows (stride 32) and mean-centered per layer before cosine geometry, plus a logit lens at each captured layer (the final norm and unembedding applied to the intermediate state). Three questions.

N1: where does the stream freeze? Per-layer trajectory geometry of the window vectors (mean step between consecutive windows, explored radius) by condition, and the logit-lens *commitment layer*, the captured layer from which the top-1 token no longer changes. Bare generation's mean step is a third to a quarter of every interrupted condition's at every sampled layer; its explored radius is 0.13 at the input layers against 0.42–0.53, and the gap narrows with depth but never closes (0.24 vs 0.33–0.34 at layer 47, CIs disjoint; Figure 14). The three interrupted conditions are geometrically alike inside the network; what separates them is what the judge reads. On the 30B, bare generation's top-1 stabilizes *later* (commitment index 11.12 $[11.04, 11.25]$ of 12 vs 10.88–10.96) while its final distribution is far more confident (final entropy 0.34 vs 0.42–1.08), a pattern consistent with copying, a late-layer computation that ends certain; the ordering does not replicate on the 8B, so we report it as an observation. The 8B and OLMo replicate the geometry (bare radius 0.14 and 0.36 at layer 0 against 0.46–0.61).

N2: which layer's movement predicts judged surprise? For every judged window, its novelty at layer l (cosine distance between the window's mean state and the mean state of

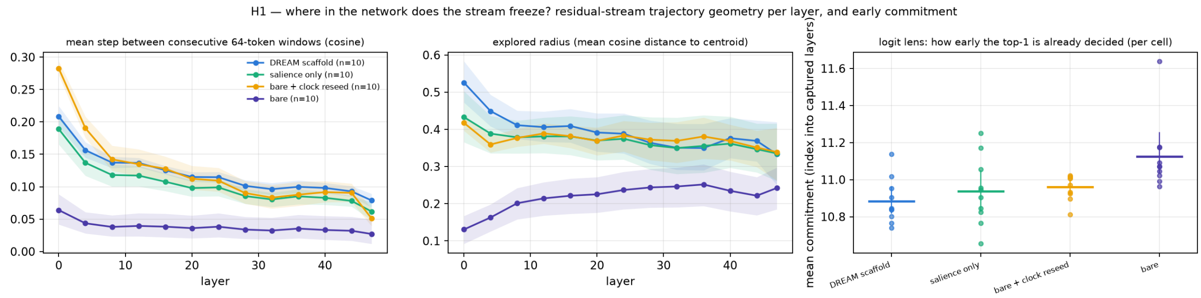

Figure 14: N1: residual-stream trajectory geometry per layer and condition (Qwen3-30B-A3B, ablation battery): mean step, explored radius, and logit-lens commitment. Bare generation moves least at every sampled layer, most so at the surface.

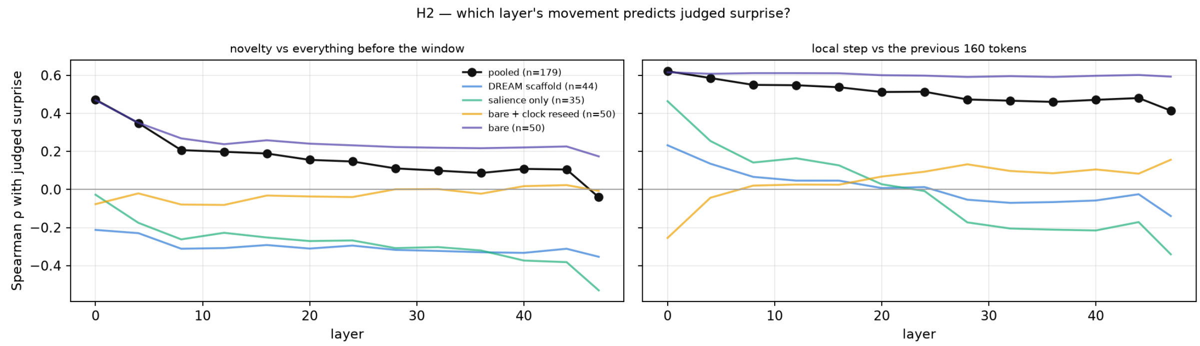

Figure 15: N2: Spearman correlation between the judged window's movement (novelty vs everything before it, left; local step, right) at each layer and judged surprise, pooled and within condition.

everything before it) and its local step (vs the previous 160 tokens), correlated with judged surprise (Figure 15). Pooled over conditions, novelty against the past correlates with surprise at the input layers ($\rho = +0.47$ at layer 0; cell-bootstrap 95% CI $[+0.22, +0.67]$) and not at the top (-0.04, [-0.26, +0.17]), but the pooled number is carried by the bare/interrupted contrast. Within interrupted conditions the sign flips with depth: local step at layer 0 correlates positively with surprise (salience-only +0.46, scaffold +0.23) and at layer 47 negatively (-0.34, -0.14); novelty against the whole past is negative at depth (salience-only -0.53 at layer 47). Across the 668 judged windows of battery 2, layer-0 movement predicts surprise within every condition (ρ +0.3 to +0.6), and higher final entropy predicts it everywhere (ρ +0.3 to +0.6): confident is unsurprising. In these data, judged surprise co-varies with departure at the surface layers and with continuity at depth: new at the surface, continuous underneath.

N3: how deep does an interruption reach? Cosine distance between the 64 tokens before an injection and the 64 after it (the injected text skipped), per layer, minus the same at random positions; and the change in similarity to the premise's own state (Figure 16). Across a scaffold reseed *with forgetting*, the state after the injection differs from the state before by +0.09 (layer 4) growing to +0.21 (layer 40) beyond control, and its similarity to the premise's state rises by +0.09 to +0.20 with depth: forgetting moves the deep state toward the premise's own state, a re-encounter with the beginning as the network represents it. Across a clock reseed *over preserved context* the same measures are +0.01–0.03 and ≈0 at every sampled layer, and this arm scores as high as any on the judged dimensions. In this reading, judged surprise and connection do not require deep representational shifts; they accompany surface departures over a deep state that is left nearly intact. We take this as a description of what the judged dimensions track, not as an account of why an arm scores as it does; the reset arm of battery 3, which by construction moves the deep state at every interruption and scores as high as the preserved context, was

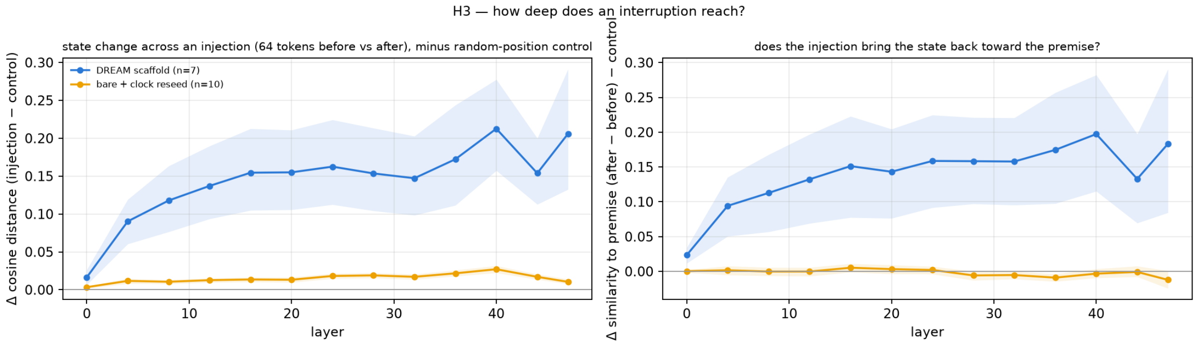

Figure 16: N3: how deep an interruption reaches (Qwen3-30B-A3B). Left: state change across an injection per layer, minus random-position control. Right: change in similarity to the premise's own state. The forgetting reseed moves the deep state and returns it toward the premise; the clock reseed over preserved context barely touches it, and scores as high on the judged dimensions.

not captured and is the obvious next probe. The dissociation replicates on the 8B (+0.03 to +0.10 vs +0.01–0.03) and, larger, on OLMo (+0.20 to +0.38 with a return of +0.13 to +0.18, vs +0.02–0.06 with no return). In battery 2, the depth of the state change scales with the length of the interrupted segment (period 75: ≈0; 150: +0.01–0.02; 300: +0.02–0.04; 600: +0.04–0.075) and not with what is injected (a dependence on segment length that the judged scores of the generated text do not show, Section 6.8), and the salience-timed stitch is the one 150-scale injection that reliably moves the deep state and pulls it toward the premise, while being the arm whose stream the judge rates lowest: deep movement and judged quality dissociate here too. Along the stream, similarity to the premise state is U-shaped in depth (≈0.9 at layer 0, ≈0.5 mid-network, rising at the top) and ordered at the top layer scaffold 0.85 > salience-only 0.79 > clock reseed 0.67 > bare 0.55: the sentence-embedding "return to the premise" of the scaffold is visible at the top layer and not in the middle of the network.

9 Discussion

The scaffold and the beam. We started from an architecture modeled on the mind (three networks, incubation, re-encounter), and measurement returned something simpler than the architecture. The DREAM scaffold works, but when it is taken apart most of its effect lives in two operations that were not the noble ones: not eating one's literal past, and being made, every few hundred tokens, to start a new sentence somewhere else. We call this minimal recipe the *interrupted loop*. Whether the earlier text is kept in the model's context while it does so turned out not to matter to the judge (Section 6.3); we kept it, and it is a choice, not an ingredient. The elaborate parts either add nothing detectable (the salience monitor as a trigger; the in-loop judge, which never fired), or interrupt too rarely (the forgetting reseed fires 0–3 times per cell, and the scaffold's lower connection tracks that rarity, not the forgetting: a complete reset every 300 tokens scores at least as well as the preserved context), or work opposite to their design (a literal return to the premise closes the loop), or do no better than the trivial alternative (salience is a good reader of the stream and, as a metronome, no better than a clock at the same rate). Only because the whole scaffold was built and then dismantled do we know which beam carries the weight, and which pieces are conditional: the reset earns its keep on a generator whose well is web boilerplate, and is indifferent on one that drifts in prose.

What an interruption is. Battery 3 takes the interruption apart as the earlier batteries took the scaffold apart. A boundary is not an interruption: a paragraph break every 300 tokens leaves the stream where habituation leaves it, and the model's own document breaks (EOS allowed) buy coherence and nothing else. A boundary that points back is worse than none. What works is a new subject; habituation adds to it (a fluent stream between injections) but the new subject alone gets most of the way. A new subject means new each time: the four-sentence rotation we used is a cycle the model closes by replaying its own earlier segments, and a window judge with a 600-token horizon reads the replay as surprise and, worse, as connection; the fresh-only numbers are the ones to believe, and a reset context, or a fresh sentence at each break, is the cure. And the new subject works whether the earlier text stays in the model's context or is thrown away, at least for what the judge can read; thrown away, it scores higher: on a reset context the model writes a fresh mini-story that returns to the premise, and the judge scores it as connected and coherent; on the preserved context it can integrate, returns less, and is entrained by thousands of tokens of its own past. So the recipe is smaller than we first wrote it: habituate, and every few hundred tokens give the model a new sentence that leads away. Keeping the memory is what a writer would want and what a 600-token judge cannot reward; the difference between integration and return is exactly the kind of long-range quality this instrument does not see, and it is where a human study with the whole document in view should look next.

Windows and wholes. The document-level judgment (Section 6.5) is the result we would least like to have and are most obliged to report: none of these streams builds anything, and the interrupted stream over the preserved context, whose windows are the best, reads as a whole below the uninterrupted habituated one. The interruption is an operator on the next hundred tokens. The architecture we started from was built for accumulation (finds kept, developed, returned to), and its one instrument for that, the judge-gated escalation, never fired; whether a gate that does open, on a real judge, can make the local gains compose is the experiment this paper did not run and the next one should.

Surface novelty and judged surprise are different quantities. The anti-probable sampler doubles n-gram novelty against the training corpus and shows no detectable effect on judged surprise or connection; the interruption lifts judged surprise by more than a point on fresh text (two to three points on all windows) without touching the sampler at all. Inside the network, read descriptively, the two live at different depths: what the judge calls surprise co-varies with lexical departure over an intact deep context, and the interruption that scores best barely moves the deep state. This is also why the neat expectation "novelty = distance from what came before" fails within interrupted streams, where deep departure from the stream's own past is negatively related to judged surprise. The reader rewards the window that is new on the surface and continuous underneath, which is what a good turn in a story is.

Rhythm, revised. The first version of this study made much of a rhythm: a break every 75 tokens develops nothing, the same break scores higher after a longer thread, and 300 tokens looked like the best period. On generated text only, most of that is gone. The post-interruption surprise does not depend on the length of the thread it breaks; on fresh windows, neither does connection; and a stream left alone for many hundreds of tokens sinks back toward the level of habituation alone. What survives is humbler and, we think, still usable: the interruption is a resource that has to be spent regularly (every 150–300 tokens on these generators), a thread needs some tens of tokens to exist before breaking it is worth anything, and nothing is gained by waiting. The coffee break of a working mind is still the right picture, provided one takes it often.

What this is not. It is not a claim that these streams are good literature; most windows are judged well below the midpoint, and the texts are what a base model produces under forced continuation. It is not a claim about creativity in the full sense (no value, interest or originality against the world is measured), nor about scale (the generators are 8–30B parameters, mostly at 8 bits), and only one post-trained model was run, in raw continuation. It is a characterization, with controls, of a simple intervention on forced open-ended generation, made on three base models from two families and under two judge families, and it points at the loop's structure and rhythm rather than at the two places where effort is usually spent.

10 Limitations

What was measured. The outcome is *judged narrative surprise, connection and coherence* on short windows of forced open-ended continuation, as read by an LLM judge and, on a subset, by human readers. That is a defensible operationalization of one ingredient of creativity, the appropriately unexpected turn, and not creativity: no value, interest or originality against the world is measured, and the study makes no claim about ideas or problems. Its title names the question the program set out with; the results answer a narrower one.

Design. Ten premises, all English narrative, one genre. Three generator models from two families for the core ladder; one generator (Qwen3-30B-A3B) for the content, timing, period and control batteries. Every arm ran once per premise with the same random seed, so the ten cells are the whole sample and a paired comparison of ten is the strongest inference available. The batteries were designed sequentially and the questions of batteries 2 and 3 were formulated after seeing battery 1 and after an external review; the analysis reported here (generated-only windows, cell as unit) was fixed before battery 3 was generated but after the earlier arms had been seen under the first protocol, so the paper should be read as a well-controlled exploratory study, not a preregistered confirmation. A confirmatory replication of the main contrast on new premises and random seeds is the obvious next step.

Instrument. The judge is one model family, calibrated for repeatability, agreeing with a second family ($\rho = 0.71$–0.85 on windows) and with the consensus of three human readers ($\rho = 0.58$ on surprise, 0.52 on coherence, on a judge-stratified sample) whose agreement with each other is low to moderate ($\alpha = 0.30$–0.38). Human readers did not rate connection in round 1, and they penalize the fluency cost of a subject change more than the judge does. The judge sees 600 tokens of context, so "connection" means connection within that horizon; text that the stream copies from further back is invisible to it as a copy, and we found (Section 6.2) that a fixed rotation of injected sentences produces exactly such copies. We report fresh-only estimates for that reason, and a judge with the whole document in view for the same reason.

Baselines and generality. The bare arm is a deliberately hard regime (forced continuation with end-of-text masked, no repetition control) and is reported as the floor of the ladder, not as a fair decoding baseline; the fair comparisons are habituation alone, habituation with a stronger penalty, EOS allowed, and the sham and reset controls. Stronger decoding baselines (look-back or contrastive decoding, learned repetition control) were not tested. Base models were used by a scope decision; the one post-trained model we ran, in raw continuation, shows the same ladder at a lower level and says nothing about the dialogue use such models are made for.

The network section. The residual-stream analysis is correlational and descriptive: mean-pooled 64-token windows at 13 sampled layers, a logit lens for a coarse commitment layer, no attention analysis, no tuned lens, no causal intervention. It characterizes what the judged dimensions correlate with in the model's state; it does not establish a mechanism.

11 Conclusion

Where does the novelty a base language model can produce with no task come from? Not, at the resolution we could reach, from the sampler: an anti-probable decoder buys surface novelty that shows no detectable effect at the level of ideas. Not, under the operationalizations we tested, from the prompt. It comes from the loop, and from a simple part of it. A base model feeding on its own output degenerates; keep it from eating its literal past and interrupt it, every few hundred tokens, with a sentence that leads away, and its generated text is judged more surprising, more connected and more coherent, by an LLM judge from another family and by human readers, on windows that contain none of the injected words. What carries it is the new subject, with or without the earlier text in the model's memory; a boundary alone does nothing; and the gains do not compose into a whole: read as documents, these streams remain collections. A pre-registered replication on new premises confirms the main contrast and refutes the one about memory. Inside the network the working interruption reads as a surface event over an intact deep state. What we have characterized is a controllable intervention on forced open-ended generation, its content, timing and period, on three base models from two families; whether the same rhythm serves a model at work on a problem, with a verifier in place of the judge, is the question this study leaves open. In this system, what an unsupervised base model can find is found in interrupting the loop.

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A Reproducibility

Code and data. The code (sampler core, MLX adapter, salience monitor, reverie engine, judges, novelty client, resumable battery runners with thermal logging, analysis and figure scripts; 117 tests), every run's token stream, per-step telemetry, event and injection positions and judgments, the human-rating packs and the dated laboratory notebook are in the repository at `https://github.com/RobertoOno/interrupting-the-loop` [public at publication; the working repository is `creative-machine`]. Judge cost of the entire program was about US$150.

Generators. `Qwen3-30B-A3B-Base` (mixture of experts, 3B active parameters), `Qwen3-8B-Base` and `OLMo-2-1124-13B`, each converted with `mlx_lm` to 8-bit affine quantization with group size 64 and run through MLX on an Apple M5 Pro (48 GB unified memory) at about 53, 31 and 17 tokens/s. Sampling is done in numpy on the full logits returned by the model. All loop arms use temperature 1.0 and the min-p floor 0.05; the anti-probable push is off ($\lambda = 0$).

Loop hyperparameters. Cell length 4,500 generated tokens; end-of-text masked except in the EOS-allowed arm. Habituation: repetition window 512 tokens, penalty factor 1.15 per occurrence (1.3 in the strong-habituation arm), applied to the log-probabilities before the floor. Interruption on a clock: a stagnation check every *period* tokens (75, 150, 300, 600 or 900) that always fires, injecting the next text of the rotation; the four neutral subject changes are listed in Section 3.3, the four return-to-the-premise stitches are *"And this, it turned out, was the same thing as the beginning, because"*, *"It came back to where it had started, of course; it always did, but this time"*, *"Which is exactly what the first line had meant, seen from here:"*, *"So the opening sentence had been true after all, only not in the way it seemed:"*; the own-past arm injects a 64-token window of the stream from at least 400 tokens back; the sham arms inject a paragraph break or *"And so, as before,"*. Reset arms rebuild the cache from the premise and the injected text only. The full scaffold: salience monitor with jump lag 32 and jump threshold 0.55, stagnation window 600 and threshold 0.08, entropy-drop trigger 0.45 after a window at or above 2.0 nats, recurrence threshold 0.15 with minimum age 96, refractory 24 steps; kick of 80 tokens ($\lambda = 4$, band $[1.2, 5.0]$, half-life 12) after a stagnation, subject change with forgetting after 3 consecutive stagnations (keeping the premise, up to 3 judged-good windows and the last 200 tokens); escalation of 120 tokens on a passing review ($\lambda = 0.5$, band $[2.5, 4.5]$); anchors every 200 tokens, at most 12; genre check every 64 tokens over 160 tokens with threshold 0.5; drift regime $\lambda = 2.5$, band $[1.8, 4.5]$, half-life 48, bridge 1.5. In the arms of this paper λ and the bridge are set to 0 in every regime. The random-number seed of the sampler is 0 in every cell.

Premises. The ten premises, shared by all arms and generators: (0) *She kept a notebook of things that had almost happened.* (1) *The map was accurate in every detail except one,* *and nobody could say which.* (2) *Every morning the baker counted the loaves twice, and every morning the count was different.* (3) *He had inherited his grandfather's watch and, with it, the habit of arriving early to places that no longer existed.* (4) *The river changed its name each time it crossed the border, and the villagers kept all the names.* (5) *The town had two clocks, and nobody remembered which one had been right first.* (6) *Her brother collected sounds the way other children collected stones.* (7) *The letter arrived forty years late and was still, somehow, on time.* (8) *Every door in the house opened onto the same room, seen from a different year.* (9) *The translator kept a list of words she refused to translate.* They were written by the author before any loop experiment and not changed afterwards.

Judge. Model `anthropic.claude-opus-5` on Amazon Bedrock (effort `high`, adaptive thinking, `max_tokens` 4000, default temperature), called through the Anthropic SDK; the event-protocol judgments were made on 14–17 August 2026 and the generated-only judgments on 18–19 August 2026. The second judge family was `moonshotai/kimi-k2.6` through OpenRouter with reasoning disabled (17 August 2026). The in-loop judge of the full scaffold was `anthropic.claude-sonnet-5` with the reverie rubric (a nearest-equivalent and novel-delta rubric whose geometric-mean score gated escalation at 5). The system prompt of the surprise judge, verbatim:

```
You judge a stretch of a language model's unsupervised reverie.  You see the
EARLIER text and the RECENT window.  Rate three INDEPENDENT things on 0-10 and
return ONLY a JSON object with keys:  "surprise" (how unexpected is the recent
window given the earlier text, a reader could not have predicted where it went;
0 = obvious continuation, 10 = genuinely startling yet not random), "connection"
(does the window bring together two distant regions of the earlier text, or an
old region with something new, in a way that makes sense; 0 if it merely continues
one thread), "coherence" (does the window hold together as text; 0 = word salad
or document boilerplate), "note" (one blunt sentence).  Rate each dimension on
its own merits; a window can be surprising and incoherent, or coherent and dull.
```

The user message is `EARLIER TEXT:\n{600 preceding tokens}\n\nRECENT WINDOW:\n{window}`. The judge is not told which condition, model or protocol produced the window.

The idea experiment. Inputs were developed by `anthropic.claude-opus-5` and judged by `anthropic.claude-sonnet-5` ($k = 3$) with the nearest-equivalent rubric; input improbability was measured as the mean cosine distance to the 10 nearest of 10,000 instructions from the Alpaca dataset (`tatsu-lab/alpaca`) embedded with `all-mpnet-base-v2`, and as perplexity under Qwen3-8B-Base.

B The human study

Recruitment, task and compensation. Raters were recruited on a freelance marketplace (Workana) with a public posting for a paid text-rating task in English; applicants were selected by the author among the platform's respondents and were not told the hypotheses, the conditions or which model produced the texts (the guide says only that the passages were written by a language model left to write on its own). Each rater received a self-contained HTML pack (rating guide embedded, English interface) with 63 windows drawn from seven conditions of the main generator, presented in a fixed random order, without any condition label, each with its preceding context; they rated surprise and coherence on the same 0–10 scales given to the model judge, at their own pace, and returned a JSON file. The guide asked them not to use AI tools or web searches for any part of the task, and raters confirmed in writing that they had not.

Compensation was R$187–300 per pack (roughly US$35–55) for an estimated 60–90 minutes of work. Round 2 (in progress): 56 generated-only windows from eight conditions (seven per condition, one per randomly chosen cell, fresh windows only, not stratified by the judge), three dimensions including connection, five raters. [results]

Ethics. The study involved adult professional raters reading machine-generated fiction; there was no deception, no sensitive content, no vulnerable population and no collection of personal data beyond the platform account used for payment. Participation was voluntary and paid; raters were informed in the posting that the texts were generated by language models and that their ratings would be used, in aggregate, in a research publication. The author is an independent researcher without access to an institutional review board; the protocol was designed to be minimal-risk and the raters remain anonymous in this paper and in the released data, which contain their ratings only. [adapt to the venue's ethics-statement requirements]

C The event-window protocol of the first version

The first version of this study cut 160-token windows at two kinds of review point, right before each injection and on a 150-token grid, so that a window after an interruption could contain the injected sentence, and treated windows as the unit. The tables of that protocol are kept below for comparison with the generated-only protocol of the main text. Across the 14 conditions of the first two batteries, the condition means under the two protocols correlate at Spearman $\rho = +0.82$ on surprise, $+0.94$ on connection and $+0.83$ on coherence (Table 21); the directions of the ladder, content and timing effects agree, the sizes differ where an injected sentence sat inside the event window, and the period and timing findings of the first version (a yield growing with the period, a stream-level advantage of the clock over salience timing) do not hold on generated text (Sections 6.7 and 6.8).

D Full tables

The tables below are generated from the run data by the analysis scripts (`scripts/analysis_gen.py` for the generated-only protocol, `analysis.py` and `analysis_b2.py` for the event protocol, `hidden_analysis.py` for the residual stream) and typeset by `scripts/appendix_tex.py`; the complete set, including every per-layer table of the residual-stream analysis, ships with the code as `docs/APPENDIX_*.md`. In the event-protocol tables the unit is the judged window (median of $k = 5$ Opus 5 judgments), windows before 100 generated tokens are excluded, and intervals are 95% bootstrap CIs over windows; they are descriptive.

Generated-only windows, unit = cell (primary analysis)

condition	n cells	surprise	connection	coherence
bare	10	0.45 [0.28, 0.68]	0.30 [0.15, 0.45]	2.52 [1.88, 3.43]
bare + habituation	10	1.60 [1.18, 2.00]	1.30 [0.83, 1.80]	4.45 [3.47, 5.47]
reseed 150, no habituation	10	2.48 [2.17, 2.78]	3.00 [2.57, 3.42]	5.37 [4.78, 5.82]
habituation + reseed 150	10	3.02 [2.53, 3.53]	3.68 [2.92, 4.52]	6.12 [5.60, 6.57]
DREAM scaffold	10	2.70 [2.10, 3.40]	1.85 [1.42, 2.33]	6.02 [5.32, 6.73]
habituation 1.3	10	1.75 [1.33, 2.22]	1.42 [1.05, 1.82]	4.88 [4.47, 5.37]
habituation, EOS allowed	10	1.52 [1.08, 1.93]	1.27 [0.93, 1.65]	5.70 [4.80, 6.47]

Table 3: Generated-only protocol. Q1: habituation × interruption (period 150) and baselines: cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
bare	surprise	-1.15 [-1.48, -0.73]	0.004	0.023	-0.80	10
bare	connection	-1.00 [-1.50, -0.48]	0.010	0.035	-0.79	10
bare	coherence	-1.93 [-3.35, -0.53]	0.035	0.070	-0.64	10
reseed 150, no habituation	surprise	+0.88 [+0.27, +1.48]	0.031	0.070	+0.70	10
reseed 150, no habituation	connection	+1.70 [+0.87, +2.43]	0.010	0.035	+0.90	10
reseed 150, no habituation	coherence	+0.92 [-0.07, +1.92]	0.135	0.187	+0.37	10
habituation + reseed 150	surprise	+1.42 [+0.92, +2.03]	0.002	0.018	+0.85	10
habituation + reseed 150	connection	+2.38 [+1.57, +3.30]	0.002	0.018	+0.93	10
habituation + reseed 150	coherence	+1.67 [+0.27, +2.88]	0.051	0.083	+0.57	10
DREAM scaffold	surprise	+1.10 [+0.40, +1.88]	0.023	0.070	+0.62	10
DREAM scaffold	connection	+0.55 [+0.15, +1.02]	0.031	0.070	+0.40	10
DREAM scaffold	coherence	+1.57 [+0.15, +2.93]	0.051	0.083	+0.57	10
habituation 1.3	surprise	+0.15 [-0.27, +0.63]	0.641	0.769	+0.01	10
habituation 1.3	connection	+0.12 [-0.43, +0.60]	0.725	0.815	+0.07	10
habituation 1.3	coherence	+0.43 [-0.70, +1.47]	0.484	0.623	+0.22	10
habituation, EOS allowed	surprise	-0.08 [-0.53, +0.37]	0.781	0.827	-0.02	10
habituation, EOS allowed	connection	-0.03 [-0.55, +0.42]	0.949	0.949	-0.05	10
habituation, EOS allowed	coherence	+1.25 [+0.07, +2.28]	0.072	0.108	+0.48	10

Table 4: Generated-only protocol. vs bare + habituation: paired by seed.

condition	n cells	surprise	connection	coherence
neutral subject change	10	3.02 [2.53, 3.53]	3.68 [2.92, 4.52]	6.12 [5.60, 6.57]
re-encounter stitch	10	2.82 [2.10, 3.55]	3.65 [2.78, 4.57]	5.52 [4.92, 6.13]
the premise itself	10	1.19 [0.89, 1.46]	1.11 [0.77, 1.43]	3.90 [3.20, 4.65]
the stream's own past	10	1.50 [1.07, 1.98]	1.42 [0.87, 2.04]	3.72 [3.07, 4.33]

Table 5: Generated-only protocol. Q2: what is injected (period 150): cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
re-encounter stitch	surprise	-0.20 [-1.15, +0.67]	0.707	0.795	-0.14	10
re-encounter stitch	connection	-0.03 [-1.40, +1.22]	0.965	0.965	-0.08	10
re-encounter stitch	coherence	-0.60 [-1.47, +0.17]	0.223	0.286	-0.42	10
the premise itself	surprise	-1.82 [-2.33, -1.32]	0.002	0.004	-1.00	10
the premise itself	connection	-2.58 [-3.42, -1.79]	0.002	0.004	-1.00	10
the premise itself	coherence	-2.22 [-3.02, -1.46]	0.002	0.004	-0.87	10
the stream's own past	surprise	-1.52 [-2.14, -0.83]	0.006	0.009	-0.84	10
the stream's own past	connection	-2.26 [-3.29, -1.29]	0.004	0.007	-0.84	10
the stream's own past	coherence	-2.40 [-3.22, -1.63]	0.002	0.004	-0.94	10

Table 6: Generated-only protocol. vs neutral subject change: paired by seed.

condition	n cells	surprise	connection	coherence
no interruption	10	1.60 [1.18, 2.00]	1.30 [0.83, 1.80]	4.45 [3.47, 5.47]
paragraph break (sham)	10	1.40 [1.07, 1.75]	0.93 [0.63, 1.25]	4.44 [3.61, 5.27]
continuity connective (sham)	10	1.08 [0.68, 1.53]	0.86 [0.47, 1.31]	3.45 [2.98, 3.97]
subject change, context preserved	10	2.92 [2.47, 3.38]	2.38 [2.03, 2.77]	6.40 [6.08, 6.70]
subject change, no habituation	10	2.47 [2.07, 2.88]	1.97 [1.55, 2.52]	5.50 [4.60, 6.33]
subject change, context reset	10	3.68 [3.37, 4.03]	3.32 [3.03, 3.62]	6.88 [6.67, 7.12]
break, context reset	10	2.03 [1.70, 2.45]	1.72 [1.48, 1.97]	5.45 [5.13, 5.83]

Table 7: Generated-only protocol. Q3: period 300: what carries the effect (context preserved vs reset; semantic change vs neutral boundary): cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
paragraph break (sham)	surprise	-0.20 [-0.70, +0.32]	0.512	0.542	-0.28	10
paragraph break (sham)	connection	-0.37 [-0.83, +0.08]	0.197	0.277	-0.28	10
paragraph break (sham)	coherence	-0.01 [-1.08, +1.22]	1.000	1.000	-0.02	10
continuity connective (sham)	surprise	-0.52 [-1.22, +0.25]	0.230	0.277	-0.41	10
continuity connective (sham)	connection	-0.44 [-1.20, +0.37]	0.330	0.371	-0.37	10
continuity connective (sham)	coherence	-1.00 [-2.35, +0.37]	0.203	0.277	-0.39	10
subject change, context preserved	surprise	+1.32 [+0.68, +1.93]	0.004	0.018	+0.85	10
subject change, context preserved	connection	+1.08 [+0.47, +1.63]	0.014	0.047	+0.74	10
subject change, context preserved	coherence	+1.95 [+0.72, +3.18]	0.021	0.055	+0.64	10
subject change, no habituation	surprise	+0.87 [+0.40, +1.35]	0.016	0.047	+0.68	10
subject change, no habituation	connection	+0.67 [-0.10, +1.48]	0.166	0.277	+0.47	10
subject change, no habituation	coherence	+1.05 [-0.13, +2.38]	0.168	0.277	+0.30	10
subject change, context reset	surprise	+2.08 [+1.68, +2.57]	0.002	0.018	+1.00	10
subject change, context reset	connection	+2.02 [+1.60, +2.40]	0.002	0.018	+0.93	10
subject change, context reset	coherence	+2.43 [+1.50, +3.33]	0.004	0.018	+0.75	10
break, context reset	surprise	+0.43 [-0.12, +1.02]	0.207	0.277	+0.38	10
break, context reset	connection	+0.42 [-0.13, +0.98]	0.219	0.277	+0.42	10
break, context reset	coherence	+1.00 [+0.13, +1.85]	0.070	0.158	+0.49	10

Table 8: Generated-only protocol. vs no interruption: paired by seed.

contrast	dim	Δ [CI]	p (perm)	Cliff δ	n
reset vs preserved (300)	surprise	+0.77 [+0.25, +1.25]	0.027	+0.61	10
reset vs preserved (300)	connection	+0.93 [+0.53, +1.32]	0.004	+0.81	10
reset vs preserved (300)	coherence	+0.48 [+0.03, +0.95]	0.090	+0.53	10
with vs without habituation (300)	surprise	+0.45 [-0.15, +1.13]	0.254	+0.36	10
with vs without habituation (300)	connection	+0.42 [-0.28, +1.07]	0.309	+0.46	10
with vs without habituation (300)	coherence	+0.90 [+0.07, +1.82]	0.117	+0.33	10
with vs without habituation (150)	surprise	+0.53 [-0.13, +1.27]	0.219	+0.32	10
with vs without habituation (150)	connection	+0.68 [-0.27, +1.67]	0.232	+0.23	10
with vs without habituation (150)	coherence	+0.75 [+0.05, +1.43]	0.080	+0.57	10
subject change vs break, both reset (300)	surprise	+1.65 [+1.13, +2.13]	0.002	+0.90	10
subject change vs break, both reset (300)	connection	+1.60 [+1.22, +2.03]	0.002	+1.00	10
subject change vs break, both reset (300)	coherence	+1.43 [+1.12, +1.77]	0.002	+0.91	10
subject change vs break, both preserved (300)	surprise	+1.52 [+0.85, +2.13]	0.006	+0.89	10
subject change vs break, both preserved (300)	connection	+1.45 [+1.03, +1.97]	0.002	+0.95	10
subject change vs break, both preserved (300)	coherence	+1.96 [+1.10, +2.86]	0.004	+0.83	10

Table 9: Generated-only protocol. Q3b: direct paired contrasts among the interruption arms.

condition	n cells	surprise	connection	coherence
stitch on the clock 150	10	2.82 [2.10, 3.55]	3.65 [2.78, 4.57]	5.52 [4.92, 6.13]
stitch on the clock 900	10	3.04 [2.70, 3.42]	2.73 [2.42, 3.07]	5.72 [5.30, 6.16]
neutral change 900	10	3.18 [2.72, 3.76]	2.28 [1.78, 2.84]	6.76 [6.58, 6.92]
stitch on salience events	10	3.04 [2.50, 3.62]	2.82 [2.23, 3.49]	5.68 [4.96, 6.47]
salience only	10	1.65 [1.23, 2.02]	1.30 [0.85, 1.77]	4.63 [3.67, 5.65]

Table 10: Generated-only protocol. Q4: timing: salience vs clock (same stitch): cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
stitch on the clock 150	surprise	-0.22 [-0.85, +0.41]	0.525	0.788	-0.22	10
stitch on the clock 150	connection	+0.92 [+0.15, +1.73]	0.074	0.223	+0.26	10
stitch on the clock 150	coherence	-0.20 [-0.72, +0.30]	0.479	0.788	-0.14	10
neutral change 900	surprise	+0.14 [-0.46, +0.82]	0.764	0.995	+0.04	10
neutral change 900	connection	-0.45 [-1.06, +0.28]	0.250	0.500	-0.45	10
neutral change 900	coherence	+1.04 [+0.60, +1.44]	0.008	0.031	+0.82	10
stitch on salience events	surprise	+0.00 [-0.73, +0.73]	1.000	1.000	-0.02	10
stitch on salience events	connection	+0.09 [-0.70, +0.83]	0.836	0.995	+0.00	10
stitch on salience events	coherence	-0.04 [-0.72, +0.65]	0.912	0.995	-0.09	10
salience only	surprise	-1.39 [-2.06, -0.83]	0.002	0.023	-0.92	10
salience only	connection	-1.43 [-2.13, -0.72]	0.006	0.031	-0.86	10
salience only	coherence	-1.09 [-2.05, +0.08]	0.100	0.239	-0.46	10

Table 11: Generated-only protocol. vs stitch on the clock 900: paired by seed.

condition	n cells	surprise	connection	coherence
stitch on the clock 900	10	2.10 [1.63, 2.52]	1.73 [1.40, 2.02]	4.78 [4.08, 5.43]
neutral change 900	10	2.51 [2.14, 2.83]	2.01 [1.68, 2.34]	5.33 [4.74, 5.88]
neutral change 600	10	2.88 [2.40, 3.32]	2.27 [1.88, 2.65]	5.50 [4.92, 6.00]
stitch on salience events	10	1.85 [1.36, 2.29]	1.78 [1.15, 2.42]	4.18 [3.74, 4.58]
DREAM scaffold	7	1.64 [1.17, 2.14]	1.46 [1.05, 1.79]	5.61 [4.65, 6.44]

Table 12: Generated-only protocol. Q4b: deep windows (at least 300 tokens after the last injection): the stream when it is left alone: cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
neutral change 900	surprise	+0.41 [-0.13, +0.95]	0.203	0.311	+0.35	10
neutral change 900	connection	+0.27 [-0.16, +0.70]	0.277	0.370	+0.25	10
neutral change 900	coherence	+0.56 [-0.14, +1.32]	0.207	0.311	+0.24	10
neutral change 600	surprise	+0.78 [+0.05, +1.47]	0.074	0.311	+0.59	10
neutral change 600	connection	+0.53 [+0.01, +1.08]	0.137	0.311	+0.44	10
neutral change 600	coherence	+0.73 [-0.12, +1.58]	0.160	0.311	+0.38	10
stitch on salience events	surprise	-0.25 [-0.80, +0.24]	0.424	0.462	-0.18	10
stitch on salience events	connection	+0.04 [-0.54, +0.64]	0.887	0.887	-0.09	10
stitch on salience events	coherence	-0.60 [-1.39, +0.18]	0.189	0.311	-0.36	10
DREAM scaffold	surprise	-0.42 [-1.10, +0.22]	0.359	0.431	-0.35	7
DREAM scaffold	connection	-0.28 [-0.56, +0.00]	0.156	0.311	-0.41	7
DREAM scaffold	coherence	+0.95 [+0.25, +1.65]	0.062	0.311	+0.43	7

Table 13: Generated-only protocol. vs stitch on the clock 900: paired by seed.

condition	n cells	surprise	connection	coherence
150	10	3.02 [2.53, 3.53]	3.68 [2.92, 4.52]	6.12 [5.60, 6.57]
300	10	2.92 [2.47, 3.38]	2.38 [2.03, 2.77]	6.40 [6.08, 6.70]
600	10	2.88 [2.62, 3.15]	1.98 [1.70, 2.30]	6.25 [5.68, 6.70]
900	10	3.18 [2.72, 3.76]	2.28 [1.78, 2.84]	6.76 [6.58, 6.92]

Table 14: Generated-only protocol. Q5: period of the neutral change: the window 32–128 tokens after the injection: cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
300	surprise	-0.10 [-0.75, +0.50]	0.832	0.832	+0.00	10
300	connection	-1.30 [-2.12, -0.57]	0.008	0.023	-0.65	10
300	coherence	+0.28 [-0.27, +0.82]	0.367	0.661	+0.22	10
600	surprise	-0.13 [-0.68, +0.35]	0.707	0.832	-0.03	10
600	connection	-1.70 [-2.43, -1.03]	0.002	0.009	-0.88	10
600	coherence	+0.13 [-0.62, +0.83]	0.766	0.832	+0.14	10
900	surprise	+0.16 [-0.25, +0.59]	0.514	0.771	+0.12	10
900	connection	-1.40 [-1.85, -0.97]	0.002	0.009	-0.64	10
900	coherence	+0.64 [+0.21, +1.08]	0.029	0.066	+0.60	10

Table 15: Generated-only protocol. vs 150: paired by seed.

period	offset 32	offset 160	offset 300	offset 450	offset 600	offset 750	stream estimate (S / C / H)
150	3.02 (n=10)	–	–	–	–	–	3.02 / 3.68 / 6.12
300	2.92 (n=10)	2.60 (n=10)	–	–	–	–	2.77 / 2.33 / 6.20
600	2.88 (n=10)	2.73 (n=10)	3.13 (n=10)	2.63 (n=10)	–	–	2.85 / 2.11 / 5.82
900	3.18 (n=10)	2.67 (n=10)	3.20 (n=10)	2.73 (n=10)	2.23 (n=10)	1.87 (n=10)	2.65 / 2.07 / 5.80

Table 16: Generated-only protocol. Q5b: decay: cell mean surprise by tokens since the injection (window start), generated text only; stream estimate = offset means weighted by the stretch of the segment each window represents.

condition	n cells	surprise	connection	coherence
bare	10	0.43 [0.22, 0.67]	0.40 [0.22, 0.60]	3.88 [2.72, 5.25]
bare + habituation	10	1.37 [1.03, 1.72]	1.00 [0.65, 1.37]	4.35 [3.35, 5.35]
habituation + reseed 150	10	2.76 [2.23, 3.22]	3.17 [2.57, 3.65]	5.15 [4.42, 5.77]
DREAM scaffold	10	2.70 [2.00, 3.43]	2.35 [1.65, 3.12]	5.38 [4.45, 6.10]

Table 17: Generated-only protocol. Q6: the ladder on Qwen3-8B-Base: cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
bare	surprise	-0.93 [-1.35, -0.53]	0.004	0.018	-0.84	10
bare	connection	-0.60 [-1.08, -0.13]	0.062	0.094	-0.61	10
bare	coherence	-0.47 [-2.13, +1.35]	0.641	0.641	-0.21	10
habituation + reseed 150	surprise	+1.39 [+0.72, +2.07]	0.008	0.018	+0.79	10
habituation + reseed 150	connection	+2.17 [+1.53, +2.83]	0.002	0.018	+0.90	10
habituation + reseed 150	coherence	+0.80 [-0.15, +1.73]	0.172	0.193	+0.28	10
DREAM scaffold	surprise	+1.33 [+0.53, +2.20]	0.016	0.028	+0.69	10
DREAM scaffold	connection	+1.35 [+0.62, +2.15]	0.008	0.018	+0.67	10
DREAM scaffold	coherence	+1.03 [-0.12, +2.17]	0.143	0.183	+0.36	10

Table 18: Generated-only protocol. vs bare + habituation: paired by seed.

condition	n cells	surprise	connection	coherence
bare	10	1.28 [0.78, 1.83]	0.93 [0.47, 1.52]	3.03 [2.18, 3.83]
bare + habituation	10	1.96 [1.30, 2.61]	1.26 [0.77, 1.76]	3.99 [2.89, 5.07]
habituation + reseed 150	10	2.77 [2.19, 3.39]	2.87 [2.30, 3.45]	4.05 [3.57, 4.52]
DREAM scaffold	10	3.60 [2.73, 4.47]	2.23 [1.60, 2.94]	6.23 [5.37, 7.05]

Table 19: Generated-only protocol. Q6: the ladder on OLMo-2-13B: cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
bare	surprise	-0.68 [-1.63, +0.26]	0.219	0.281	-0.40	10
bare	connection	-0.33 [-1.11, +0.45]	0.484	0.545	-0.25	10
bare	coherence	-0.96 [-2.33, +0.12]	0.184	0.275	-0.29	10
habituation + reseed 150	surprise	+0.82 [-0.21, +1.81]	0.166	0.275	+0.36	10
habituation + reseed 150	connection	+1.61 [+0.83, +2.35]	0.004	0.035	+0.82	10
habituation + reseed 150	coherence	+0.06 [-0.99, +1.20]	0.938	0.938	+0.01	10
DREAM scaffold	surprise	+1.64 [+0.42, +2.97]	0.043	0.097	+0.53	10
DREAM scaffold	connection	+0.97 [+0.26, +1.84]	0.023	0.070	+0.48	10
DREAM scaffold	coherence	+2.24 [+0.87, +3.63]	0.020	0.070	+0.68	10

Table 20: Generated-only protocol. vs bare + habituation: paired by seed.

condition	surprise event / gen	connection event / gen	coherence event / gen	n cells (event / gen)
abl_forget	3.44 / 2.72	1.73 / 1.90	3.88 / 6.13	10 / 10
salience only	2.86 / 1.65	1.53 / 1.30	3.85 / 4.63	10 / 10
bare	0.28 / 0.45	0.20 / 0.30	2.06 / 2.52	10 / 10
bare + habituation	1.70 / 1.60	1.08 / 1.30	4.08 / 4.45	10 / 10
habituation + reseed 150	3.08 / 3.02	3.15 / 3.68	4.78 / 6.12	10 / 10
reseed 300	4.01 / 2.92	3.12 / 2.38	5.98 / 6.40	10 / 10
reseed 600	3.64 / 2.88	2.64 / 1.98	5.50 / 6.25	10 / 10
reseed 900	4.28 / 3.18	2.53 / 2.28	5.55 / 6.76	10 / 10
stitch 900	3.91 / 3.04	3.01 / 2.73	5.17 / 5.72	10 / 10
premise 150	1.02 / 1.19	0.98 / 1.11	3.03 / 3.90	10 / 10
re-encounter stitch 150	2.55 / 2.82	3.53 / 3.65	4.67 / 5.52	10 / 10
own past 150	1.37 / 1.50	1.50 / 1.42	3.03 / 3.72	10 / 10
salience-timed stitch	3.00 / 3.04	2.26 / 2.82	3.85 / 5.68	10 / 10
DREAM scaffold	3.50 / 2.70	1.70 / 1.85	3.82 / 6.02	10 / 10

Table 21: Generated-only protocol. Protocol agreement: condition means, event windows (v1) vs generated-only windows (unit = cell).

condition	n windows	copied	copied, source outside the judge's 600 tokens	fresh windows: S / C / H (cell means, n cells)	copied windows: S / C / H
bare	60	67%	0%	1.77 / 1.38 / 5.55 (n=10)	0.03 / 0.00 / 1.45
bare + habituation	60	27%	8%	1.76 / 1.30 / 5.06 (n=10)	0.94 / 1.00 / 2.69
habituation 1.3	60	18%	3%	1.96 / 1.54 / 5.38 (n=10)	0.64 / 0.73 / 2.91
habituation, EOS allowed	60	15%	3%	1.52 / 1.30 / 5.88 (n=10)	1.22 / 1.11 / 3.67
reseed 150, no habituation	60	80%	7%	2.93 / 1.83 / 6.07 (n=10)	2.40 / 3.27 / 5.29
habituation + reseed 150	60	72%	12%	2.95 / 2.03 / 6.78 (n=10)	3.09 / 4.28 / 6.19
reseed 300, no habituation	60	68%	63%	2.40 / 1.65 / 5.55 (n=10)	2.51 / 2.12 / 5.51
reseed 300	60	65%	62%	3.07 / 2.03 / 6.52 (n=10)	2.82 / 2.56 / 6.36
reseed 600	60	38%	35%	2.85 / 2.04 / 6.33 (n=10)	2.91 / 2.00 / 6.30
reseed 900	50	12%	12%	3.10 / 2.27 / 6.73 (n=10)	4.17 / 2.67 / 7.00
re-encounter stitch 150	60	68%	45%	3.38 / 2.98 / 6.33 (n=10)	2.68 / 3.83 / 5.29
stitch 900	50	10%	6%	3.20 / 2.80 / 5.83 (n=10)	2.20 / 2.60 / 4.60
premise 150	60	60%	7%	2.31 / 1.68 / 5.05 (n=10)	0.64 / 0.81 / 3.07
own past 150	60	72%	27%	2.50 / 1.37 / 4.98 (n=10)	1.24 / 1.50 / 3.28
salience-timed stitch	30	30%	27%	3.10 / 3.12 / 5.68 (n=9)	3.67 / 3.00 / 5.22
paragraph break 300 (sham)	60	32%	5%	2.08 / 1.29 / 4.92 (n=10)	0.21 / 0.53 / 3.00
continuity connective 300 (sham)	60	73%	3%	3.42 / 1.99 / 5.87 (n=10)	0.34 / 0.41 / 2.68
reset + new subject 300	60	0%	0%	3.68 / 3.32 / 6.88 (n=10)	–
reset + break 300	60	0%	0%	2.03 / 1.72 / 5.45 (n=10)	–
DREAM scaffold	31	6%	0%	2.69 / 1.92 / 6.24 (n=10)	2.50 / 1.50 / 2.00

Table 22: Generated-only protocol. Self-copy: verbatim reproduction of the stream's own earlier text inside the judged windows.

generator	condition	n windows	copied	fresh windows: S / C / H (n cells)
Qwen3-8B-Base	bare	60	63%	1.15 / 1.11 / 6.70 (n=9)
Qwen3-8B-Base	bare + habituation	60	45%	2.53 / 1.90 / 6.03 (n=10)
Qwen3-8B-Base	habituation + reseed 150	60	78%	3.48 / 2.07 / 5.77 (n=10)
Qwen3-8B-Base	DREAM scaffold	26	8%	2.78 / 2.41 / 5.47 (n=10)
OLMo-2-13B	bare	60	45%	2.11 / 1.51 / 4.89 (n=10)
OLMo-2-13B	bare + habituation	60	20%	2.20 / 1.33 / 4.31 (n=10)
OLMo-2-13B	habituation + reseed 150	60	45%	3.19 / 2.57 / 3.89 (n=10)
OLMo-2-13B	DREAM scaffold	28	4%	3.60 / 2.23 / 6.48 (n=10)

Table 23: Generated-only protocol. Self-copy on the other generators (ladder arms; copied = >= 50% shingles seen earlier in the stream).

contrast	dim	Δ [CI]	p (perm)	Cliff δ	n seeds
interruption 150 vs habituation	surprise	+1.19 [+0.42, +2.09]	0.010	+0.74	10
interruption 150 vs habituation	connection	+0.73 [+0.37, +1.15]	0.008	+0.52	10
interruption 150 vs habituation	coherence	+1.72 [+0.35, +2.86]	0.041	+0.63	10
interruption 300 vs habituation	surprise	+1.30 [+0.73, +1.87]	0.004	+0.87	10
interruption 300 vs habituation	connection	+0.73 [+0.33, +1.22]	0.008	+0.64	10
interruption 300 vs habituation	coherence	+1.45 [+0.43, +2.44]	0.023	+0.64	10
reset + subject change 300 vs habituation	surprise	+1.92 [+1.59, +2.26]	0.002	+1.00	10
reset + subject change 300 vs habituation	connection	+2.02 [+1.73, +2.33]	0.002	+1.00	10
reset + subject change 300 vs habituation	coherence	+1.82 [+1.07, +2.59]	0.004	+0.75	10
reset vs preserved (300)	surprise	+0.62 [+0.10, +1.13]	0.070	+0.43	10
reset vs preserved (300)	connection	+1.28 [+0.82, +1.73]	0.004	+0.93	10
reset vs preserved (300)	coherence	+0.37 [-0.10, +0.82]	0.188	+0.34	10
scaffold vs habituation	surprise	+0.93 [+0.21, +1.71]	0.062	+0.53	10
scaffold vs habituation	connection	+0.62 [+0.16, +1.12]	0.055	+0.39	10
scaffold vs habituation	coherence	+1.17 [-0.02, +2.38]	0.102	+0.50	10
interruption 900 vs habituation	surprise	+1.33 [+0.81, +1.78]	0.004	+0.93	10
interruption 900 vs habituation	connection	+0.97 [+0.37, +1.57]	0.020	+0.73	10
interruption 900 vs habituation	coherence	+1.67 [+0.73, +2.61]	0.012	+0.72	10

Table 24: Generated-only protocol. Fresh windows only: paired contrasts (cells with at least one fresh window).

condition	n	integration	development	coherence	surprise
bare	10	0.90 [0.50, 1.30]	0.50 [0.20, 0.80]	1.10 [0.60, 1.60]	0.60 [0.20, 1.00]
bare + habituation	10	1.40 [0.90, 1.90]	1.10 [0.60, 1.70]	1.50 [1.20, 1.80]	1.40 [1.10, 1.70]
habituation + reseed 150	10	0.70 [0.30, 1.10]	0.30 [0.00, 0.60]	1.20 [1.00, 1.50]	0.10 [0.00, 0.30]
reseed 300	10	0.80 [0.50, 1.00]	0.40 [0.10, 0.70]	1.50 [1.20, 1.80]	0.50 [0.20, 0.80]
reseed 300, no habituation	10	0.80 [0.40, 1.20]	0.20 [0.00, 0.50]	1.40 [1.10, 1.70]	0.40 [0.10, 0.70]
paragraph break 300 (sham)	10	1.40 [0.80, 2.00]	1.00 [0.60, 1.40]	1.50 [0.90, 2.10]	1.00 [0.60, 1.40]
reset + new subject 300	10	2.00 [1.60, 2.40]	1.60 [1.30, 1.90]	2.20 [1.80, 2.60]	2.50 [2.20, 2.80]
reset + break 300	10	0.80 [0.40, 1.20]	0.40 [0.10, 0.70]	1.00 [1.00, 1.00]	0.80 [0.40, 1.20]
DREAM scaffold	10	1.10 [0.80, 1.40]	1.10 [0.80, 1.40]	1.00 [0.70, 1.30]	1.30 [1.00, 1.60]
salience only	10	1.20 [0.80, 1.60]	1.00 [0.60, 1.50]	1.50 [1.20, 1.80]	1.30 [1.00, 1.60]
reseed 600	10	1.70 [1.40, 2.00]	0.90 [0.70, 1.00]	1.70 [1.40, 2.00]	1.20 [1.00, 1.50]
reseed 75	10	0.60 [0.30, 0.90]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]	0.10 [0.00, 0.30]
reseed 900	10	1.70 [1.40, 2.00]	1.20 [1.00, 1.50]	1.70 [1.30, 2.10]	1.50 [1.20, 1.80]
stitch 900	10	1.80 [1.40, 2.20]	1.10 [1.00, 1.30]	1.90 [1.60, 2.20]	1.60 [1.20, 2.00]
premise 150	10	0.60 [0.10, 1.20]	0.40 [0.10, 0.80]	1.10 [0.80, 1.40]	0.40 [0.10, 0.80]
re-encounter stitch 150	10	1.30 [1.00, 1.60]	0.30 [0.00, 0.60]	1.40 [1.10, 1.70]	0.40 [0.10, 0.70]
own past 150	10	0.40 [0.10, 0.70]	0.10 [0.00, 0.30]	1.00 [0.70, 1.30]	0.20 [0.00, 0.50]
salience-timed stitch	10	1.70 [1.40, 2.00]	0.80 [0.50, 1.00]	1.50 [1.20, 1.80]	0.90 [0.60, 1.20]

Table 25: Generated-only protocol. Document level: the whole 4,500-token stream, injected sentences removed, Opus k=3 (180 documents).

contrast	dim	Δ [CI]	p (perm)	Cliff δ	n
interruption 150 vs habituation	integration	-0.70 [-1.50, +0.10]	0.203	-0.46	10
interruption 150 vs habituation	development	-0.80 [-1.50, -0.30]	0.062	-0.56	10
interruption 150 vs habituation	coherence	-0.30 [-0.60, +0.00]	0.250	-0.30	10
interruption 150 vs habituation	surprise	-1.30 [-1.60, -1.00]	0.002	-0.94	10
interruption 300 (preserved) vs habituation	integration	-0.60 [-1.10, -0.10]	0.109	-0.42	10
interruption 300 (preserved) vs habituation	development	-0.70 [-1.40, -0.10]	0.125	-0.48	10
interruption 300 (preserved) vs habituation	coherence	+0.00 [-0.30, +0.30]	1.000	+0.00	10
interruption 300 (preserved) vs habituation	surprise	-0.90 [-1.40, -0.40]	0.031	-0.70	10
reset + subject change 300 vs habituation	integration	+0.60 [+0.10, +1.10]	0.109	+0.42	10
reset + subject change 300 vs habituation	development	+0.50 [-0.10, +1.00]	0.234	+0.42	10
reset + subject change 300 vs habituation	coherence	+0.70 [+0.20, +1.20]	0.062	+0.55	10
reset + subject change 300 vs habituation	surprise	+1.10 [+0.70, +1.50]	0.008	+0.80	10
reset vs preserved (300)	integration	+1.20 [+0.70, +1.60]	0.008	+0.84	10
reset vs preserved (300)	development	+1.20 [+0.70, +1.60]	0.008	+0.84	10
reset vs preserved (300)	coherence	+0.70 [+0.30, +1.10]	0.031	+0.55	10
reset vs preserved (300)	surprise	+2.00 [+1.50, +2.50]	0.002	+1.00	10
scaffold vs habituation	integration	-0.30 [-0.70, +0.00]	0.500	-0.20	10
scaffold vs habituation	development	+0.00 [-0.60, +0.50]	1.000	+0.06	10
scaffold vs habituation	coherence	-0.50 [-0.80, -0.20]	0.062	-0.45	10
scaffold vs habituation	surprise	-0.10 [-0.30, +0.00]	1.000	-0.10	10
sham break vs habituation	integration	+0.00 [-0.60, +0.50]	1.000	+0.02	10
sham break vs habituation	development	-0.10 [-0.50, +0.30]	1.000	-0.02	10
sham break vs habituation	coherence	+0.00 [-0.60, +0.60]	1.000	+0.05	10
sham break vs habituation	surprise	-0.40 [-0.70, -0.10]	0.125	-0.32	10
habituation vs bare	integration	+0.50 [+0.00, +1.00]	0.180	+0.32	10
habituation vs bare	development	+0.60 [+0.20, +1.10]	0.125	+0.40	10
habituation vs bare	coherence	+0.40 [-0.20, +1.00]	0.406	+0.35	10
habituation vs bare	surprise	+0.80 [+0.20, +1.40]	0.078	+0.60	10

Table 26: Generated-only protocol. Document level: paired contrasts.

condition	n cells	surprise	connection	coherence
no interruption	7	1.95 [0.60, 3.60]	1.79 [0.40, 3.67]	4.86 [3.31, 6.24]
subject change, context preserved	8	3.10 [2.50, 3.79]	2.96 [2.42, 3.52]	6.19 [5.56, 6.73]
paragraph break (sham)	9	1.70 [1.00, 2.43]	1.19 [0.54, 2.00]	5.20 [3.91, 6.35]
subject change, no habituation	9	2.22 [1.56, 2.81]	2.02 [1.37, 2.63]	5.48 [4.35, 6.57]
subject change, context reset	8	4.29 [3.56, 5.10]	3.46 [2.75, 4.31]	6.21 [5.40, 6.88]

Table 27: Generated-only protocol. Confirmatory: period 300 on new premises: cell means (mean over cells [95% CI over cells]).

condition	dim	Δ [CI]	p (perm)	q (BH)	Cliff δ	n seeds
subject change, context preserved	surprise	+1.94 [+1.03, +2.72]	0.062	0.250	+0.75	6
subject change, context preserved	connection	+2.11 [+1.53, +2.75]	0.031	0.188	+0.86	6
subject change, context preserved	coherence	+2.08 [+0.72, +3.61]	0.031	0.188	+0.78	6
paragraph break (sham)	surprise	-0.14 [-2.78, +1.81]	0.969	1.000	+0.08	6
paragraph break (sham)	connection	-0.67 [-3.64, +1.69]	0.750	1.000	+0.19	6
paragraph break (sham)	coherence	+0.22 [-2.61, +2.78]	0.906	1.000	+0.19	6
subject change, no habituation	surprise	+0.05 [-1.79, +1.69]	1.000	1.000	+0.22	7
subject change, no habituation	connection	+0.00 [-1.81, +1.62]	1.000	1.000	+0.31	7
subject change, no habituation	coherence	+0.19 [-1.02, +1.21]	0.750	1.000	+0.02	7
subject change, context reset	surprise	+2.47 [+0.67, +3.60]	0.125	0.300	+0.76	5
subject change, context reset	connection	+1.90 [+0.67, +2.70]	0.125	0.300	+0.76	5
subject change, context reset	coherence	+2.33 [+0.23, +4.33]	0.188	0.375	+0.68	5

Table 28: Generated-only protocol. vs no interruption: paired by seed.

hypothesis	contrast	dim	Δ (mean of paired differences) [CI]	p	n seeds
H1 (primary)	clock300 vs bare_habit	surprise	+1.94 [+1.03, +2.72]	0.0312	6
H2	clock300 vs sham_break300	surprise	+0.71 [-0.00, +1.29]	0.0547	7
H3	clock300 vs reset_reseed300	connection	-0.43 [-1.57, +0.55]	0.7344	7
H4 (two-sided)	clock300 vs nohabit300	surprise	+1.05 [+0.07, +2.12]	0.1875	7

Table 29: Generated-only protocol. Pre-registered contrasts (exact one-sided sign-flip permutation unless stated).

condition	n	integration	development	coherence	surprise
bare + habituation	10	1.40	0.90	1.50	1.10
paragraph break 300 (sham)	10	0.60	0.40	1.40	1.10
reseed 300	10	1.20	0.40	1.30	0.60
reseed 300, no habituation	10	0.80	0.10	1.20	0.10
reset + new subject 300	10	2.10	1.60	1.90	2.50

Table 30: Generated-only protocol. Confirmatory premises: document level (50 documents).

contrast	dim	Δ [CI]	p (perm)	n
preserved vs habituation	integration	-0.20 [-0.90, +0.50]	0.797	10
preserved vs habituation	development	-0.50 [-1.00, +0.10]	0.234	10
preserved vs habituation	coherence	-0.20 [-0.60, +0.20]	0.625	10
preserved vs habituation	surprise	-0.50 [-1.20, +0.20]	0.312	10
reset vs habituation	integration	+0.70 [+0.10, +1.20]	0.094	10
reset vs habituation	development	+0.70 [+0.30, +1.10]	0.031	10
reset vs habituation	coherence	+0.40 [+0.00, +0.80]	0.219	10
reset vs habituation	surprise	+1.40 [+0.90, +1.90]	0.004	10
reset vs preserved	integration	+0.90 [+0.40, +1.40]	0.031	10
reset vs preserved	development	+1.20 [+0.80, +1.60]	0.004	10
reset vs preserved	coherence	+0.60 [+0.30, +0.90]	0.031	10
reset vs preserved	surprise	+1.90 [+1.50, +2.30]	0.002	10

Table 31: Generated-only protocol. Confirmatory premises: document level (50 documents).

Event-window protocol of the first version (descriptive)

dimension	condition	n	mean ± sd	scaffold − cond CI95	Cliff's δ	Mann-Whitney p
surprise	DREAM scaffold (λ=0)	47	3.19 ± 1.92	–	–	–
surprise	salience only	38	2.63 ± 1.75	[-0.22, +1.33]	+0.16	0.195
surprise	bare + clock reseed	60	3.08 ± 1.48	[-0.54, +0.78]	+0.00	0.967
surprise	bare generation	50	0.28 ± 0.72	[+2.34, +3.50]	+0.88	4.58e-15
connection	DREAM scaffold (λ=0)	47	1.64 ± 1.56	–	–	–
connection	salience only	38	1.39 ± 1.42	[-0.40, +0.87]	+0.09	0.446
connection	bare + clock reseed	60	3.15 ± 1.53	[-2.09, -0.91]	-0.58	1.52e-07
connection	bare generation	50	0.20 ± 0.57	[+0.98, +1.93]	+0.65	7.08e-10
coherence	DREAM scaffold (λ=0)	47	3.68 ± 1.69	–	–	–
coherence	salience only	38	3.71 ± 1.60	[-0.72, +0.67]	-0.03	0.819
coherence	bare + clock reseed	60	4.78 ± 0.95	[-1.65, -0.56]	-0.42	0.00014
coherence	bare generation	50	2.06 ± 2.01	[+0.87, +2.35]	+0.57	5.34e-07

Table 32: Ablation battery (Qwen3-30B-A3B, 10 seeds, Opus 5 k=5): scaffold vs each condition, unit = window.

dimension	windows	mean spread	p90 spread
surprise	242	0.61	1.00
connection	242	0.60	1.00
coherence	242	0.68	1.00

Table 33: Instrument calibration: intra-window spread of k=5 judgments per dimension (Opus 5).

arm	n	novel 4-grams %	novel 6-grams %	Δ4 vs baseline CI95	Cliff's δ	max copied span (words)
min-p baseline	15	21.5 ± 10.1	70.9 ± 11.9	–	–	10
λ=1	15	36.0 ± 12.4	87.6 ± 7.4	[+0.069, +0.230]	+0.68	10
λ=2	15	45.5 ± 10.8	89.7 ± 5.9	[+0.163, +0.314]	+0.88	13

Table 34: Phase 1: objective novelty against the OLMo-2 training corpus (3 prompts × 5 seeds).

arm	runs	train-plateau escapes	mean champion test excess	best
plain	5	2/5	0.0602	0.0600
antiprob	5	3/5	0.0604	0.0601

Table 35: Verified search on Qwen3-30B-A3B (bin packing): the anti-probable operator does not separate from plain sampling.

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
bare (no habituation, no interruption)	50	jump	0.28 ± 0.72	0.20 ± 0.57	2.06 ± 2.01	0/50
bare + habituation (no interruption)	60	clock	1.70 ± 1.71	1.08 ± 1.10	4.08 ± 2.19	4/60
bare + habituation + clock reseed 150	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
DREAM scaffold (λ=0)	47	crystallize/jump/recurrence	3.19 ± 1.92	1.64 ± 1.56	3.68 ± 1.69	8/47

Table 36: Battery 2, Q1: habituation and interruption (Qwen3-30B-A3B).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
bare (no habituation, no interruption)	surprise	-1.42 [-1.89, -0.95]	-0.58	1.4e-08	-1.42 [-1.72, -1.12] (10)
bare (no habituation, no interruption)	connection	-0.88 [-1.20, -0.56]	-0.52	1.1e-07	-0.88 [-1.22, -0.54] (10)
bare (no habituation, no interruption)	coherence	-2.02 [-2.79, -1.23]	-0.55	3.4e-07	-2.02 [-3.37, -0.68] (10)
bare + habituation + clock reseed 150	surprise	+1.38 [+0.80, +1.93]	+0.50	1.3e-06	+1.38 [+0.92, +1.92] (10)
bare + habituation + clock reseed 150	connection	+2.07 [+1.60, +2.55]	+0.75	5.2e-13	+2.07 [+1.63, +2.57] (10)
bare + habituation + clock reseed 150	coherence	+0.70 [+0.10, +1.30]	+0.21	4.3e-02	+0.70 [-0.27, +1.72] (10)
DREAM scaffold (λ=0)	surprise	+1.49 [+0.80, +2.19]	+0.46	3.4e-05	+1.80 [+0.97, +2.75] (10)
DREAM scaffold (λ=0)	connection	+0.55 [+0.06, +1.10]	+0.20	6.2e-02	+0.61 [+0.22, +1.01] (10)
DREAM scaffold (λ=0)	coherence	-0.40 [-1.14, +0.34]	-0.10	3.8e-01	-0.26 [-1.03, +0.56] (10)

Table 37: Battery 2, Q1: habituation and interruption (Qwen3-30B-A3B); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
neutral subject change (clock 150)	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
re-encounter stitch (clock 150)	50	cut	2.68 ± 1.46	3.64 ± 1.57	4.72 ± 1.33	5/50
the premise itself (clock 150)	50	cut	1.14 ± 1.73	1.04 ± 1.57	3.18 ± 1.90	3/50
the stream's own past (clock 150)	50	cut	1.42 ± 1.43	1.52 ± 1.35	3.06 ± 1.42	2/50

Table 38: Battery 2, Q2: content of the interruption at period 150.

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
re-encounter stitch (clock 150)	surprise	-0.40 [-0.95, +0.14]	-0.13	2.4e-01	-0.40 [-1.29, +0.46] (10)
re-encounter stitch (clock 150)	connection	+0.49 [-0.09, +1.07]	+0.20	7.1e-02	+0.49 [-0.56, +1.56] (10)
re-encounter stitch (clock 150)	coherence	-0.06 [-0.50, +0.38]	-0.00	9.8e-01	-0.06 [-0.75, +0.57] (10)
the premise itself (clock 150)	surprise	-1.94 [-2.53, -1.31]	-0.68	6.1e-10	-1.94 [-2.65, -1.30] (10)
the premise itself (clock 150)	connection	-2.11 [-2.68, -1.51]	-0.72	4.1e-11	-2.11 [-2.83, -1.46] (10)
the premise itself (clock 150)	coherence	-1.60 [-2.15, -1.01]	-0.53	1.2e-06	-1.60 [-2.36, -0.70] (10)
the stream's own past (clock 150)	surprise	-1.66 [-2.21, -1.11]	-0.61	2.1e-08	-1.66 [-2.26, -1.07] (10)
the stream's own past (clock 150)	connection	-1.63 [-2.17, -1.08]	-0.62	9.0e-09	-1.63 [-2.37, -0.91] (10)
the stream's own past (clock 150)	coherence	-1.72 [-2.18, -1.24]	-0.70	1.1e-10	-1.72 [-2.30, -1.11] (10)

Table 39: Battery 2, Q2: content of the interruption at period 150; differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
re-encounter on the clock (150)	50	cut	2.68 ± 1.46	3.64 ± 1.57	4.72 ± 1.33	5/50
re-encounter on the clock (900; matched frequency)	40	cut	2.73 ± 1.95	2.30 ± 1.82	4.67 ± 1.59	6/40
neutral change on the clock (900; matched frequency)	40	cut	2.48 ± 1.87	2.02 ± 1.68	4.95 ± 2.16	5/40
re-encounter on salience events (event windows)	80	crystallize/cut/jump/recurrence	3.92 ± 1.42	2.99 ± 1.73	3.75 ± 1.18	8/80
re-encounter on salience events (uniform windows)	60	clock	2.18 ± 2.04	1.68 ± 1.65	3.93 ± 1.88	4/60
re-encounter on the clock (900), uniform windows	60	clock	4.70 ± 1.69	3.48 ± 1.52	5.50 ± 1.41	31/60
neutral change on the clock (900), uniform windows	60	clock	5.48 ± 1.64	2.87 ± 1.56	5.95 ± 1.32	45/60
salience only, no injection (event windows)	38	crystallize/jump/recurrence	2.63 ± 1.75	1.39 ± 1.42	3.71 ± 1.60	2/38

Table 40: Battery 2, Q3: timing of the re-encounter, clock vs salience.

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
re-encounter on the clock (900; matched frequency)	surprise	+0.04 [-0.68, +0.78]	-0.03	8.1e-01	+0.04 [-0.53, +0.66] (10)
re-encounter on the clock (900; matched frequency)	connection	-1.34 [-2.03, -0.62]	-0.45	2.2e-04	-1.34 [-2.17, -0.53] (10)
re-encounter on the clock (900; matched frequency)	coherence	-0.04 [-0.65, +0.58]	-0.02	8.9e-01	-0.05 [-0.87, +0.73] (10)
neutral change on the clock (900; matched frequency)	surprise	-0.21 [-0.90, +0.51]	-0.10	4.4e-01	-0.21 [-0.84, +0.36] (10)
neutral change on the clock (900; matched frequency)	connection	-1.62 [-2.28, -0.92]	-0.51	2.5e-05	-1.61 [-2.17, -1.13] (10)
neutral change on the clock (900; matched frequency)	coherence	+0.23 [-0.56, +0.99]	+0.11	3.5e-01	+0.23 [-0.45, +1.07] (10)
re-encounter on salience events (event windows)	surprise	+1.24 [+0.73, +1.75]	+0.44	2.0e-05	+1.10 [+0.34, +1.85] (10)
re-encounter on salience events (event windows)	connection	-0.65 [-1.23, -0.06]	-0.24	2.2e-02	-0.92 [-1.55, -0.29] (10)
re-encounter on salience events (event windows)	coherence	-0.97 [-1.41, -0.52]	-0.39	1.5e-04	-0.86 [-1.72, -0.06] (10)
re-encounter on salience events (uniform windows)	surprise	-0.50 [-1.14, +0.15]	-0.21	5.5e-02	-0.50 [-1.11, -0.01] (10)
re-encounter on salience events (uniform windows)	connection	-1.96 [-2.54, -1.35]	-0.61	2.7e-08	-1.96 [-2.62, -1.37] (10)
re-encounter on salience events (uniform windows)	coherence	-0.79 [-1.38, -0.19]	-0.25	2.1e-02	-0.79 [-1.31, -0.29] (10)
re-encounter on the clock (900), uniform windows	surprise	+2.02 [+1.43, +2.60]	+0.64	5.9e-09	+2.02 [+1.62, +2.44] (10)
re-encounter on the clock (900), uniform windows	connection	-0.16 [-0.74, +0.43]	-0.03	7.7e-01	-0.16 [-0.68, +0.37] (10)
re-encounter on the clock (900), uniform windows	coherence	+0.78 [+0.29, +1.30]	+0.31	4.4e-03	+0.78 [+0.40, +1.15] (10)
neutral change on the clock (900), uniform windows	surprise	+2.80 [+2.23, +3.37]	+0.77	1.9e-12	+2.80 [+2.04, +3.59] (10)
neutral change on the clock (900), uniform windows	connection	-0.77 [-1.37, -0.18]	-0.27	1.3e-02	-0.77 [-1.78, +0.23] (10)
neutral change on the clock (900), uniform windows	coherence	+1.23 [+0.74, +1.73]	+0.48	7.2e-06	+1.23 [+0.56, +1.84] (10)
salience only, no injection (event windows)	surprise	-0.05 [-0.73, +0.64]	-0.05	7.0e-01	+0.18 [-0.80, +1.31] (10)
salience only, no injection (event windows)	connection	-2.25 [-2.87, -1.62]	-0.73	3.8e-09	-2.11 [-3.00, -1.16] (10)
salience only, no injection (event windows)	coherence	-1.01 [-1.64, -0.37]	-0.35	4.5e-03	-0.87 [-1.92, +0.12] (10)

Table 41: Battery 2, Q3: timing of the re-encounter, clock vs salience; differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
every 75 (before injection)	50	cut	0.86 ± 1.04	1.28 ± 1.04	3.08 ± 0.63	1/50
every 150 (before injection)	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
every 300 (before injection)	50	cut	2.90 ± 1.50	2.70 ± 1.63	5.88 ± 0.82	7/50
every 600 (before injection)	50	cut	3.06 ± 1.58	2.56 ± 1.66	5.32 ± 1.62	6/50
every 900 (before injection)	40	cut	2.48 ± 1.87	2.02 ± 1.68	4.95 ± 2.16	5/40
every 300 (mid-segment)	60	clock	4.93 ± 1.53	3.47 ± 1.68	6.07 ± 0.98	34/60
every 600 (mid-segment)	60	clock	4.12 ± 1.83	2.70 ± 1.63	5.65 ± 1.54	23/60
every 900 (mid-segment)	60	clock	5.48 ± 1.64	2.87 ± 1.56	5.95 ± 1.32	45/60

Table 42: Battery 2, Q4: frequency of interruption (neutral change).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
every 75 (before injection)	surprise	-2.22 [-2.69, -1.75]	-0.83	1.7e-14	-2.22 [-2.91, -1.73] (10)
every 75 (before injection)	connection	-1.87 [-2.35, -1.38]	-0.76	1.1e-12	-1.87 [-2.49, -1.32] (10)
every 75 (before injection)	coherence	-1.70 [-2.00, -1.41]	-0.86	1.5e-15	-1.70 [-2.00, -1.39] (10)
every 300 (before injection)	surprise	-0.18 [-0.74, +0.37]	-0.06	5.7e-01	-0.18 [-0.81, +0.39] (10)
every 300 (before injection)	connection	-0.45 [-1.04, +0.12]	-0.18	1.1e-01	-0.45 [-1.31, +0.45] (10)
every 300 (before injection)	coherence	+1.10 [+0.78, +1.42]	+0.59	2.5e-08	+1.10 [+0.69, +1.57] (10)
every 600 (before injection)	surprise	-0.02 [-0.60, +0.54]	-0.00	9.8e-01	-0.02 [-0.74, +0.60] (10)
every 600 (before injection)	connection	-0.59 [-1.19, +0.01]	-0.21	4.9e-02	-0.59 [-1.28, +0.03] (10)
every 600 (before injection)	coherence	+0.54 [+0.03, +1.03]	+0.26	1.8e-02	+0.54 [-0.09, +1.13] (10)
every 900 (before injection)	surprise	-0.61 [-1.28, +0.08]	-0.21	6.7e-02	-0.61 [-1.41, +0.18] (10)
every 900 (before injection)	connection	-1.12 [-1.78, -0.47]	-0.38	1.1e-03	-1.12 [-2.11, -0.26] (10)
every 900 (before injection)	coherence	+0.17 [-0.56, +0.88]	+0.12	3.1e-01	+0.17 [-0.79, +1.12] (10)
every 300 (mid-segment)	surprise	+1.85 [+1.30, +2.38]	+0.60	9.0e-09	+1.85 [+1.23, +2.33] (10)
every 300 (mid-segment)	connection	+0.32 [-0.27, +0.88]	+0.13	2.0e-01	+0.32 [-0.30, +0.77] (10)
every 300 (mid-segment)	coherence	+1.28 [+0.93, +1.62]	+0.63	1.1e-09	+1.28 [+0.88, +1.68] (10)
every 600 (mid-segment)	surprise	+1.03 [+0.45, +1.62]	+0.33	1.3e-03	+1.03 [+0.58, +1.47] (10)
every 600 (mid-segment)	connection	-0.45 [-1.03, +0.10]	-0.17	9.1e-02	-0.45 [-1.00, +0.17] (10)
every 600 (mid-segment)	coherence	+0.87 [+0.40, +1.32]	+0.42	4.6e-05	+0.87 [+0.32, +1.43] (10)
every 900 (mid-segment)	surprise	+2.40 [+1.83, +2.93]	+0.70	1.5e-11	+2.40 [+1.98, +2.78] (10)
every 900 (mid-segment)	connection	-0.28 [-0.83, +0.27]	-0.09	3.8e-01	-0.28 [-0.75, +0.10] (10)
every 900 (mid-segment)	coherence	+1.17 [+0.75, +1.57]	+0.56	7.2e-08	+1.17 [+0.78, +1.57] (10)

Table 43: Battery 2, Q4: frequency of interruption (neutral change); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

every	windows	pooled surprise	conn	coh	early (≤160 tokens since injection) S / C / H	late (>160) S / C / H	phase-weighted stream mean S / C / H
75	60	0.88	1.35	3.15	0.88 / 1.35 / 3.15 (n=60)	, /: /: (n=0)	0.88 / 1.35 / 3.15
150	60	3.08	3.15	4.78	3.08 / 3.15 / 4.78 (n=60)	, /: /: (n=0)	3.08 / 3.15 / 4.78
300	110	4.01	3.12	5.98	5.28 / 3.35 / 5.95 (n=40)	3.29 / 2.99 / 6.00 (n=70)	4.35 / 3.18 / 5.97
600	110	3.64	2.64	5.50	5.40 / 3.20 / 5.65 (n=20)	3.24 / 2.51 / 5.47 (n=90)	3.82 / 2.69 / 5.52
900	100	4.28	2.53	5.55	5.90 / 2.80 / 6.10 (n=40)	3.20 / 2.35 / 5.18 (n=60)	3.68 / 2.43 / 5.35

Table 44: Battery 2, Q4b: phase within the segment and phase-weighted stream means.

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
8B: bare	110	clock/jump	0.68 ± 1.26	0.38 ± 0.73	3.45 ± 2.64	1/110
8B: bare + habituation	60	clock	1.47 ± 1.69	0.87 ± 1.23	3.93 ± 2.26	4/60
8B: bare + habituation + clock reseed 150	60	clock/cut	2.73 ± 1.41	3.23 ± 1.60	4.42 ± 1.28	1/60
8B: DREAM scaffold (λ=0)	146	clock/crystallize/cut/jump	2.73 ± 2.13	1.53 ± 1.47	4.30 ± 1.88	15/146

Table 45: Second generator family (Qwen3-8B-Base).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
8B: bare	surprise	-0.78 [-1.28, -0.30]	-0.30	3.0e-04	-0.78 [-1.13, -0.45] (10)
8B: bare	connection	-0.48 [-0.84, -0.16]	-0.27	7.1e-04	-0.48 [-0.84, -0.14] (10)
8B: bare	coherence	-0.49 [-1.23, +0.28]	-0.13	1.5e-01	-0.49 [-2.43, +1.58] (10)
8B: bare + habituation + clock reseed 150	surprise	+1.26 [+0.69, +1.80]	+0.47	6.3e-06	+1.26 [+0.52, +2.14] (10)
8B: bare + habituation + clock reseed 150	connection	+2.37 [+1.87, +2.87]	+0.80	9.4e-15	+2.37 [+1.68, +3.12] (10)
8B: bare + habituation + clock reseed 150	coherence	+0.48 [-0.18, +1.13]	+0.13	2.1e-01	+0.48 [-0.53, +1.43] (10)
8B: DREAM scaffold (λ=0)	surprise	+1.27 [+0.70, +1.81]	+0.38	1.8e-05	+1.17 [+0.62, +1.68] (10)
8B: DREAM scaffold (λ=0)	connection	+0.66 [+0.25, +1.04]	+0.31	2.7e-04	+0.67 [+0.27, +1.07] (10)
8B: DREAM scaffold (λ=0)	coherence	+0.36 [-0.30, +1.01]	+0.10	2.4e-01	+0.52 [-0.44, +1.27] (10)

Table 46: Second generator family (Qwen3-8B-Base); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
OLMo: bare	109	clock/jump	1.35 ± 1.80	0.91 ± 1.53	2.97 ± 2.65	6/109
OLMo: bare + habituation	59	clock	2.37 ± 1.91	1.51 ± 1.50	4.05 ± 2.68	7/59
OLMo: bare + habituation + clock reseed 150	60	clock/cut	3.11 ± 1.63	3.17 ± 1.54	3.28 ± 1.63	6/60
OLMo: DREAM scaffold (λ=0)	144	clock/crystallize/cut/jump	3.08 ± 1.94	1.89 ± 1.54	4.48 ± 2.32	22/144

Table 47: Third generator family (OLMo-2-13B).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
OLMo: bare	surprise	-1.02 [-1.62, -0.44]	-0.36	7.8e-05	-1.02 [-2.08, -0.10] (10)
OLMo: bare	connection	-0.60 [-1.07, -0.13]	-0.30	4.8e-04	-0.60 [-1.52, +0.31] (10)
OLMo: bare	coherence	-1.08 [-1.92, -0.26]	-0.25	6.1e-03	-1.08 [-2.34, -0.01] (10)
OLMo: bare + habituation + clock reseed 150	surprise	+0.74 [+0.09, +1.36]	+0.26	1.5e-02	+0.72 [-0.30, +1.53] (10)
OLMo: bare + habituation + clock reseed 150	connection	+1.66 [+1.10, +2.20]	+0.58	3.0e-08	+1.65 [+0.77, +2.30] (10)
OLMo: bare + habituation + clock reseed 150	coherence	-0.77 [-1.58, +0.03]	-0.12	2.4e-01	-0.78 [-1.77, +0.25] (10)
OLMo: DREAM scaffold (λ=0)	surprise	+0.71 [+0.13, +1.29]	+0.22	1.4e-02	+0.81 [+0.18, +1.47] (10)
OLMo: DREAM scaffold (λ=0)	connection	+0.38 [-0.09, +0.84]	+0.17	5.6e-02	+0.49 [-0.02, +1.02] (10)
OLMo: DREAM scaffold (λ=0)	coherence	+0.43 [-0.34, +1.20]	+0.11	2.3e-01	+0.69 [-0.33, +1.63] (10)

Table 48: Third generator family (OLMo-2-13B); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	n	radius L0	radius L12	radius L24	radius L36	radius L47	commit layer (idx)	final entropy
DREAM scaffold	10	0.526	0.406	0.388	0.350	0.333	10.88 [10.82, 10.96]	1.08
salience only	10	0.433	0.381	0.375	0.355	0.335	10.94 [10.83, 11.05]	0.88
bare + clock reseed	10	0.418	0.389	0.383	0.381	0.339	10.96 [10.92, 11.00]	0.42
bare	10	0.131	0.214	0.237	0.252	0.242	11.12 [11.04, 11.25]	0.34

Table 49: Residual-stream geometry (Qwen3-30B-A3B, ablation battery): explored radius per captured layer (subset), stable commitment layer index and final entropy, means over cells.